Reliable Face Ethnicity and Gender Inference

Anonymous WACV Datasets Track submission

Paper ID 985

Abstract

Fairness evaluation in face analysis systems (FAS) uses automatic demographic attribute inference (DAI), which in turn relies on predefined demographic segmentation. However, the validity of fairness auditing hinges on the reliability of the DAI process. We revisit this process by proposing a fully reproducible ethnicity+gender prediction pipeline that includes curation of diversified training and test datasets and a transfer-learning-inspired model training strategy. We integrate pretrained face recognition encoders with non-linear classification heads, which we prove leverage the encoders' intrinsic latent structure for attribute inference. We audit this pipeline across three dimensions: accuracy, fairness, and a newly introduced notion of robustness, defined via intra-identity consistency. We compare our approach with seven competitive baselines across multiple datasets and training setups, including pipelines trained specifically for attribute prediction and generalist VLMs. Our pipeline outperforms all baselines for ethnicity, the more challenging attribute, across the three audited dimensions. It also ensures a very interesting performance-efficiency balance for gender. Importantly, the newly proposed attribute training dataset generalizes better than existing counterparts, while the test dataset is more challenging. An ablation study assesses the main pipeline components. To promote reproducibility and adoption, we release the new dataset metadata, full codebase, pretrained models, and evaluation toolkit.

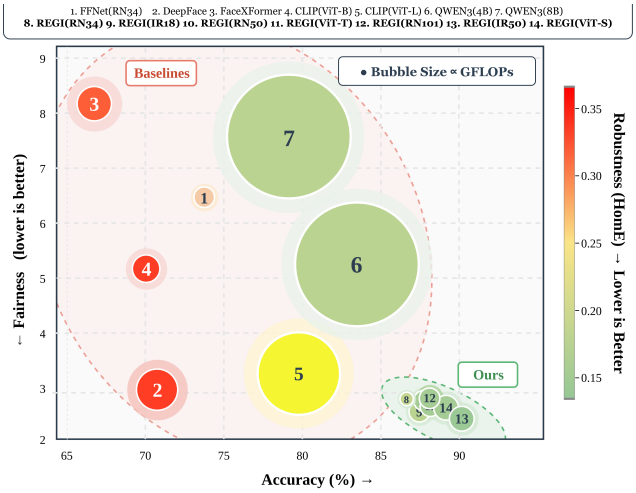


Figure 1. Ethnicity inference results (4 classes) audited using accuracy, fairness, and robustness dimensions on our proposed WEG^t dataset for baselines (FairFace [34], the DeepFace Library [64], FaceXFormer [47]), and pretrained VLMs [7, 55] and several instances of the proposed approach. Higher values are better for accuracy. **Lower** values are better for **fairness**. **Greener** is better for robustness. The circle sizes are proportional to the number of GFLOPs, i.e. number of floating-point operations per image inference, except for QWEN3, which would be too large to fit in the figure. Smaller circles indicate better efficiency. An ideal DAI system would be represented by a collapsed circle located at the bottom-right corner of the plot.

1. Introduction

The accuracy of automatic face analysis systems (FAS) has significantly progressed over the past decade, driven by increasingly sophisticated models [16, 36] and the availability of large-scale training datasets [82]. FAS now powers critical applications, from smartphone authentication to national security infrastructure [6, 68, 74]. Given this societal impact, it must be not only accurate but also fair to ensure responsible deployment and public trust [57]. Numerous studies have shown that FAS performance can vary significantly across demographic groups [9, 21, 25, 26, 37, 62]. These disparities are attributable to demographic imbalance in training data [26, 49, 78], limited data diversity [21], visual variability [72], or suboptimal training procedures [14, 33, 38]. While these studies have advanced our understanding of algorithmic bias, they largely overlook a foundational assumption: the reliability of the demographic attribute inference (DAI) process itself. In practice, attribute inference typically relies on off-the-shelf tools [37], such as FairFace pipeline [34], referred to as FFNet to avoid any confusion with the FairFace dataset referred as such, the

DeepFace library [64], or VLMs [7]. Yet, as illustrated in Figure 1, these tools offer suboptimal performance and compromise the reliability of downstream attribute analysis. We thus revisit ethnicity and gender inference and show that significant performance gains can be achieved by carefully curating DAI datasets and replacing the standard end-to-end learning strategy. We predict four coarse ethnicities (*African*, *Asian*, *Caucasian*, *Indian*) and two genders (*Female*, *Male*) for comparability with existing systems and datasets.

First, we propose **WEG**, a large curated **E**thnicity and **G**ender dataset that resamples **W**ebFace [82] while emphasizing demographic diversity. We infer both attributes with FFNet [34] and aggregate predictions at the identity level to improve prediction reliability. We boost diversity by matching WebFace identities to a set of seed identities that represent a wide range of ethnicities within each coarse category. We also ensure gender balance in the proposed test and training subsets, and impose non-contamination by separating test and training identities.

Second, we introduce reliable gender and ethnicity inferenc (**REGI**), a theoretically grounded and reliable DAI (ethnicity and gender) approach that establishes a theoretical bridge between probing strategy and cos similarity base encoder. Previous approaches based on end-to-end training [34, 47] or full fine-tuning [64] build upon the hypothesis that they outperform probing, an instantiation of transfer-learning [66]. There is, nevertheless, no theoretical consensus on this in a general case and, in the particular case of demographic attribute analysis, as shown for age [50]. We rather pretrain face recognition backbones and then adapt a classification head for ethnicity and gender recognition. Despite such strategy is similar to older schemes, we notably find that using an RBF-SVM classifier, which provides a non-decision boundary well-suited to angular representation spaces, outperforms linear classifiers, end-to-end training, and fine-tuning, as well as the MLP introduced for age estimation in classification in [50]. We provide a theoretical justification for this behavior for embeddings trained with angular losses, which are common in face recognition [16, 36, 75].

Third, we advocate for conducting ethnicity and gender auditing across three complementary dimensions: **accuracy, fairness, and robustness**. Accuracy underpins correct interpretations; fairness ensures equitable error distribution across groups; robustness, an underexplored property, promotes intra-identity consistency. We introduce a novel robustness metric based on the simple yet sound hypothesis: *An ideal demographic classifier should predict the same segment for all images of the same person.* We formalize this metric through an entropy-based robustness metric that generalizes across demographic taxonomies.

We validate these contributions through extensive experiments across various backbones, attribute training datasets, pretraining datasets, and classification heads. We illustrate our experimental findings for ethnicity inference in Figure 1. REGI simultaneously improves accuracy, fairness, and robustness compared with widely used systems, namely the FFNet model [34], released alongside the FairFace dataset, DeepFace [64], and FaceXFormer [47], as well as large pretrained VLMs, such as CLIP [55] or Qwen3 [7], while also being more efficient.

We improve the reliability, transparency, and reproducibility of ethnicity and gender inference. These three dimensions are critical for valid fairness evaluations in downstream face-related tasks. The use of publicly available datasets, not always respected in existing systems such as FFNet [34] or Qwen3 [7], ensures transparency. The release of the full, packaged code (attached in the supplementary materials) for training and inference will facilitate reproducibility. We hope REGI will serve as a reliable foundation for future developments that further improve DAI accuracy, fairness, and robustness. To reduce the risk of misuse, we will release the demographic inference tool under a restricted-use license (see appendix code) limited to fairness auditing and related research. Authorized uses include aggregate bias measurement and mitigation within models. The license explicitly forbids individual-level profiling, identification, authentication, surveillance, targeting, or decision-making, as well as any deployment for biometric categorisation outside the context of lawful, aggregate fairness assessment.

2. Related Work

Fairness evaluation [9, 21, 25, 26, 56, 62] is needed to comprehensively audit AI-driven face analysis systems deployed in practical applications, including border control [29], consumer authentication [71], public safety [45], face-based payment [59]. These studies highlight significant performance variation across demographic segments, due to imbalance in face recognition training datasets [26, 78], insufficient data diversity [21], or biases in the architectures or training strategies [19, 46, 58]. While understanding and mitigating biases is important, little effort is devoted to the reliability of the demographic attribute inference that enables their conclusions. Below, we analyze the main components of existing inference pipelines, as well as other relevant works.

The demographic segmentation schemes represent the basis for any demographic classifier. They are all criticizable because they categorize the vast diversity of human faces into a limited set of categories [37]. For instance, a widely used scheme [34, 64] splits faces into coarse, predefined ethnic categories, such as *African, Asian, Caucasian* and *Indian*. Alternatives were proposed, notably based on skin color [11], color and shape [44], or tone [41, 73], but

their semantic interpretation is less direct compared to that of predefined categories. Importantly, a preliminary experiment reported in Appendix E highlights much lower intra-identity consistency for color-based splits compared with ethnicity-based categorization. Consequently, we adopt the latter scheme in our work.

Training datasets are a core component of demographic classification. In supervised settings, their annotation follows a predefined segmentation. Representative examples include RFW [76], BFW [60], FairFace [34], and BUPT [77] datasets. Their common characteristics include the manual curation of demographic annotations and the emphasis on segment balancing. While balancing usually increases fairness, multiple studies [3, 21, 38, 62] show that it only partially addresses DAI unfairness. The authors of [72] investigate the contribution of non-demographic factors on face recognition biases. Different works adapted the training process to improve fairness: SensitiveLoss [65], a discrimination-aware loss, penalizes demographic disparities, GAC [24] adapts convolution kernels and attention mechanisms to faces based on their demographic attributes, and DebFace [23] uses adversarial learning to reduce face recognition and DAI biases jointly, while LAOFIW [4] tackles attribute combinations during training. The authors of [78] examine the dataset structure (number of identities and images per identity) and factors related to DAI fairness, such as head pose, brightness, and image quality. These studies inform our experiments, and we test different attribute training dataset structures and sizes.

Ethnicity and gender auditing most often employs models trained specifically for the task. FFNet, the model released alongside FairFace [34] is a widely used demographic classification tool [13, 18, 21, 39, 43, 62] that jointly infers ethnicity, gender, and age. The authors demonstrate that training on their balanced dataset yields more equitable performance across demographic groups compared to models trained on imbalanced alternatives. One important limitation is their released model (FFNet) 4-ethnicity classification accuracy (88% on the RFW and BFW datasets), which affects the relevance of fairness auditing based on it. Beyond performance limitations, FFNet suffers from a critical reproducibility gap: The authors have not released training code or sufficient implementation details to enable independent replication, despite numerous attempts [1], including ours. This opacity undermines scientific reproducibility, which is essential for bias auditing, leaving practitioners unable to verify claimed performance or to adapt the methodology to new contexts. FaceXFormer [47] trains a transformer architecture on a merge of FFNet [34] and UTKFace [80] for demographic attribute inference. The DeepFace library [64] takes a different approach, and fully finetunes a pretrained model to infer face attributes. General-purpose VLMs leverage large multimodal datasets and can be applied to DAI, among other tasks. Their zero-shot capabilities make them an appealing alternative to models trained specifically for the task. However, VLMs are less efficient due to the large size of their backbones, as illustrated in Figure 1.

[1] https://github.com/joojs/fairface/issues

3. Ethnicity and gender attribute inference

We illustrate different instantiations of demographic inference pipelines in Figure 2, highlighting in green our contributions: **(1)** a diversified training set sampled from WebFace42M [82] after reliable majority voting for demographic labeling, **(2)** a transfer-learning-inspired training that advantageously combined pretrained face recognition encoders and a non-linear classification head, **(3)** a challenging test set also sampled from WebFace42M, **(4)** a new robustness metric that quantifies intra-identity prediction consistency.

3.1. Ethnicity and gender datasets (WEG)

Figure 2 provides a generic view of the DAI pipeline. A curated dataset includes a set $S = s_1, s_2, ..., s_n$ of demographic segments (combination of Ethnicity+Gender). Ethnicity is defined using the predefined categories *African, Asian, Caucasian, Indian* [34, 77], used here for comparability with the previous DAI systems [34, 47, 64]. Alternative encodings based on country of origin [40, 54], or skin appearance [44, 73] could nevertheless be used. We encode gender using the *female* and *male* segments. Any segmentation scheme introduces ethical and technical challenges [26, 35, 54], as it reduces the diversity of human faces into a finite set of n classes. In practice, we argue that the segmentation scheme should be selected to jointly optimize classification accuracy, fairness, and robustness. Beyond the segmentation choice itself, the dataset should include diverse, sufficiently large samples for each segment, with reliable, high-quality labels [77]. Since labeling is expensive, we combine ethnicity and gender inference in the WebFace42M face recognition dataset [82], leveraging intra-identity robustness (see subsection 3.3 and subsection E.3) to consolidate demographic labels. Furthermore, we aim to sample a dataset from WebFace 42M that enforces diversification not only across the 4 coarse ethnicity categories but also across multiple fine-grained ethnicities. To this end, we use an auxiliary seed dataset whose images are not part of the proposed datasets but are used to drive diversification.

Seed dataset. We initiated the collection with a list of 270 fine-grained ethnicities and then prompted GPT-5.1 to generate up to 100 male and 100 female names for each ethnicity. The identities must refer to living people or, at a minimum, be active in the 21st century. The collected metadata

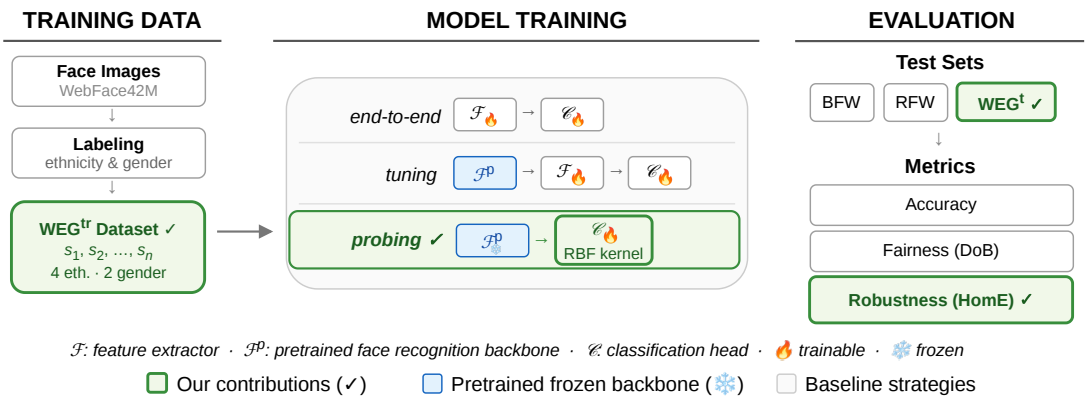


Figure 2. Pipeline for training and evaluating demographic attribute inference (s_i - ethnicity + gender segments ($n = 8$); $\mathcal{F}$ - feature extractor; $\mathcal{F}\checkmark$ - pretrained $\mathcal{F}$, $\mathcal{C}$ - classification head). We highlight our contributions in green. The training dataset is annotated with 4 ethnicity and 2 gender labels (thus 8 segments). The model training component implements different strategies. We compare our probing strategy with end-to-end DAI training [34, 47] and full fine-tuning [64]. We propose a diversified test set and add robustness evaluation to the classical analysis of accuracy and fairness.

includes a Wikipedia page for each identity, if available. Following this listing and filtering of generated entries, 60 images per name were collected and filtered using a method inspired by [10]: Each selected face must, on the one hand, be closer to a prototypical identity than to any other prototypical identity in the database, and, on the other hand, be considered similar to other faces of the same identity. We manually verified image filtering for 100 identities, achieving 99.3% accuracy.

After filtering, we mapped each identity into the four ethnicities and two gender categories: FFNet infers attributes for individual images and aggregates them at the identity level using a majority vote.

We establish thresholds for identity filtering using 500 ethnicity+gender labeled BUPT identities [77], and evaluate the MiMA robustness metrics (see Table 6) for gender and ethnicity attributions, achieving correct rates of 98.12% and 90%, respectively. We note that fine-grained ethnicities are social constructs that mix cultural and origin-based dimensions [20], and that individuals can be assigned to different coarse-grained categories. This yields 244, 164, 241, and 211 for the African, Asian, Caucasian, and Indian fine categories. The gender distribution is 12k/15k between females and males. This dataset is not perfectly balanced, but it remains sufficiently varied as a seed for resampling diverse identities from WebFace. The details of the seed dataset, are provided in Appendix D.

WebFace resampling. We randomly split the 2M identities in the labeled WebFace42M into 80%, 10%, and 10% for training, validation, and test datasets, abbreviated WEG^{tr}, WEG^{v}, and WEG^{t}, respectively. This prior separation ensures there is no identity contamination between the three subsets. A coarse identity and gender information is inferred for each identity following the same algorithm and thresholds as for the seed dataset. Image embeddings for both datasets are extracted using an AdaFace model [36] pretrained on WebFace42M. We associate each WebFace identity with the closest seed. A WebFace identity is compared to a seed identity by averaging the minimum distance for each image of this WebFace identity to the seed identity images. The minimal mean distance designates the proxy for the WebFace identity. We filter out the IDs from WebFace whose coarse ethnicity or gender differs from their assigned proxy seed id. Furthermore, we retain only identities containing at least 20 images and for which gender consistency exceeds 75%, and ethnicity 60%. This filtering leads to a subset of WebFace composed by 135, 67, 172, and 79 fine ethnicities for African, Asian, Caucasian, and Indian categories. We emphasize that identity associations are only used as proxies for diversification and should not be regarded as explicit ethnicity attribution.

We diversify WEG^{v} and WEG^{t} by sampling uniformly among coarse ethnicities, and then among fine ethnicities that compose them, based on the distribution of seed identities across fine-grained ethnicities within each coarse category. For creating an efficient training set we follow a different diversification process: Let M and N be the number of fine and coarse ethnicities, $p(g|e)$ denote the conditional probability of group g (Asian, Indian, etc.) given ethnicity subgroup e, satisfying $\sum_{g=1}^{N} p(g|e) = 1$ for all e. We denote by $p^*(.|e)$ the closest probability distribution from $p(.|e)$ satisfying for all g $S_g = \frac{M}{N}$ where $S_g = \sum_{e=1}^{M} p(g|e)$ denotes the total mass of group g across all subgroups. We propose the following two-stage sampling procedure: (1) Sample a group g uniformly with probabil-

ity $P(g) = \frac{1}{N}$ and (2) Within group g, sample subgroup e with probability $p^*(.|e)$. A formal justification and description of this sampling is given in Appendix A as well as a discussion about the associated labels.

3.2. Model training (REGI)

Widely used DAI models ($\mathcal{M}$), comprising a feature extractor $\mathcal{F}$ and a classifier $\mathcal{C}$, still rely on training from scratch [34, 47] or full fine-tuning [64] of pretrained models. This persists despite the proven benefits of transfer learning in related face analysis tasks [5, 21, 48, 69]. Motivated by these findings, we explore the use of pretrained models $\mathcal{F}^p$ as an alternative. In the probing setting (Figure 2), we train the classifier $\mathcal{C}$ directly on top of a frozen backbone $\mathcal{F}^p$. Prior DAI works [30, 32] that employed pretraining typically started from ImageNet-pretrained models, often with linear heads. More specifically, a common assumption [34, 62] is that the demographic segments S are linearly separable in the embedding space, motivating the use of a linear classifier $\mathcal{C}$. We, however, challenge this assumption and derive an approach that combines two highly compatible backbone and head: a backbone pretrained with an angular loss, and an RBF-SVM.

Theoretical Justification for using RBF-SVM. We provide a novel unified geometric and probabilistic framework that theoretically justifies the use of RBF kernel classifiers for transfer learning from face recognition models pretrained with angular margin losses to demographic attribute inference. To our knowledge, it is the first end-to-end theoretical analysis connecting (i) von Mises-Fisher (vMF)-structured pretraining, (ii) vMF kernel similarity, and (iii) vMF mixture decision boundaries in a unified framework for this transfer learning scenario. Specifically, we characterize RBF-SVM decision functions as comparisons between von Mises-Fisher mixture models—an interpretation that has not appeared in prior work and provides principled guidance for architectural choices for gender and ethnicity inference in fairness-aware face recognition. The main results and contribution are outlined below and detailed in Appendix C.

Modern face recognition models employ angular margin losses [16, 36, 75], which operate on L2-normalized feature embeddings in $\mathbb{R}^d$, thus lies on the unit hypersphere $\mathbb{S}^{d-1}$. The von Mises-Fisher (vMF) distribution is the canonical probability distribution for directional data on the unit hypersphere [42]. It is already known [42, 63] that the softmax classifier with L2-normalized features and class centers is equivalent to a generative model where each class follows a vMF distribution. The angular margins modify this by effectively decreasing the likelihood of the correct class during training, forcing the model to achieve higher concentration (smaller intra-class angles) to overcome the penalty. This results in compact, well-separated vMF-like identity clusters on $\mathbb{S}^{d-1}$ after convergence.

We retrieve a classical result from directional statistics [42], namely that *the RBF kernel on normalized features computes a similarity measure proportional to the vMF likelihood*. Then, our key theoretical contribution consists in **characterizing the RBF-SVM decision function as a comparison between vMF mixture models**. This interpretation has not appeared in prior work and provides new insight into why RBF kernels are theoretically suited for this transfer learning scenario.

Let us consider binary demographic classification where $y_i \in \{-1, +1\}$ indicates class membership. The SVM decision function is:

$$f(\mathbf{z}) = \sum_{i \in \mathcal{S}} \alpha_i y_i K_{\text{RBF}}(\mathbf{z}_i, \mathbf{z}) + b, \tag{1}$$

We show that one can define two vMF mixture densitie p_+ and p_- from the support vectors, such that the decision boundary is reformulated as:

$$f(\mathbf{z}) = \Lambda \left[\beta_+ p_+(\mathbf{z}) - \beta_- p_-(\mathbf{z})\right] + b, \tag{2}$$

where $\beta_\pm$ represent the total mass assigned to each class and Λ, b are some constants. We further provide an interpretation of this results that shows it approximates a log-likelihood ratio test:

$$\hat{y} = \text{sign}\left[\log \frac{p_+(\mathbf{z})}{p_-(\mathbf{z})}\right] = \text{sign}\left[\log p_+(\mathbf{z}) - \log p_-(\mathbf{z})\right], \tag{3}$$

which is the optimal Bayes classifier for choosing between two generative models.

Implications: Geometric Consistency for Transfer Learning This theoretical analysis reveals a fundamental geometric consistency in our transfer learning pipeline that has not been previously formalized:

1. **Pretraining with angular margin losses** (e.g., AdaFace) are known to organize face embeddings on the hypersphere such that same-identity samples form tight vMF-like clusters, while different identities are angularly separated [16, 63].
2. **RBF kernel SVMs** operating on these normalized embeddings implicitly leverage the vMF geometry, providing non-linear decision boundaries that respect the spherical structure-the kernel-vMF connection is established [42], but its application to this transfer learning scenario is novel.
3. **Decision boundaries as vMF mixture comparisons** (Equation 3) provide a new probabilistic interpretation: the RBF-SVM learns to discriminate between demographic attributes by comparing which vMF mixture (constructed from support vectors) better explains a test

embedding: **this characterization is our novel contribution and the full theoretical derivation can be found in Appendix C.**

3.3. Robustness as complementary evaluation dimension

DAI evaluation typically involves accuracy metrics, often complemented by fairness indicators [62]. This design allows per-segment behavior analysis and deeper performance diagnostics [21]. A key challenge is to balance accuracy and fairness, as improvements in one may degrade the other [79]. We include both in our evaluation and introduce robustness as a complementary dimension.

We posit that a reliable attribute classifier should produce consistent predictions for different images of the same identity. In practice, this consistency is not guaranteed due to variations in illumination, pose, resolution, and other capture conditions [44, 73], as illustrated in Figure 1.

We define a robustness metric below that is **taxonomy-agnostic**, since it leverages intra-identity consistency. Let us consider an image dataset $\mathcal{I} = \{I_j^{id}; id \in [\![1, D]\!], j \in [\![1, N_{id}]\!]\}$, with D identities, an identity id being represented by N_{id} images. We consider a labeling function $f_{\mathcal{A}}^{\mathcal{C}}$ that attributes a label to each image, each label being an element of a finite set $\mathcal{C}$, and depends on a dataset $\mathcal{A}$ ($f_{\mathcal{A}}^{\mathcal{C}}$ is defined below). For each identity id, we compute the relative frequency of each possible label $k \in \mathcal{C}$ noted as $p_{id}^k = \frac{1}{N_{id}} \sum_{j=1}^{N^{id}} \delta_{f_{\mathcal{A}}^{\mathcal{C}}(I_j^{id}) = f_{\mathcal{A}}^{\mathcal{C}}(I_k^{id})}$. We then compute the **homogeneity entropy** as the average entropy of p over all identities:

$$HomE = \frac{1}{D} \sum_{id} \frac{1}{\log(C)} \sum_{k} -p_{id}^k \log(p_{id}^k) \qquad (4)$$

This metric captures intra-identity consistency. Lower values of HomE indicate higher robustness and are thus desirable for trustworthy DAI pipelines.

4. Experiments

We first present the experimental setup used to audit the inference of demographic attributes. Next, we report and analyze the main results obtained with both the baseline and the proposed DAI pipelines. Finally, we present ablation experiments to study the contribution of the main components of our approach.

4.1. Experimental setup

Baselines. We compare our probing-based demographic attribute inference approach against competitive baselines, including both task-specific and generalist VLMs with this capability. The first category includes: (1) the FairFace pipeline FFNet that implements end-to-end training from Fig. 2 using ResNet34 (RN34) [28] and combines a publicly released DAI subset with unavailable proprietary data [34]. (2) The DeepFace library [64][2] that fully finetunes a VGGFace model [51] pretrained for face recognition, and combines data from multiple DAI sources, and (3) the recent FaceXFormer model [47] that implements multi-task end-to-end training of a Swin-B model with data sourced from UTKFace [80] and FairFace [47]. The second category includes a CLIP model [55] with ViT-B and ViT-L backbones and a QWEN3-VL-Instruct [7] with 4B and 8B parameters, respectively. These models represent a representative subset of current approaches for predicting ethnicity and gender.

Test datasets. We evaluate the reliability of ethnicity and gender inference using publicly available datasets selected based on the availability of demographic attributes and identity labels. BFW [61], and WEG^t employ a common four-class ethnicity and two-gender segmentation and are used for accuracy and fairness evaluation. They include multiple samples per identity, enabling robustness evaluation. RFW [76] only includes ethnicity information per sample, and no identity information. Consequently, we use it only to evaluate the accuracy and fairness of ethnicity predictions.

Metrics. We report micro-average for accuracy evaluation. For fairness, we use the degree of bias (DoB), computed as the standard deviation of per-group accuracies (lower is better). For robustness, we use the Homogeneity Entropy (HomE) metric defined in Eq. (4), along with additional robustness measures described in Appendix E.

Ethnicity and gender inference pipelines. In Table 1, we fix the pretraining dataset (28k identities and 1.4M WebFace42M samples balanced across ethnicity and gender), the classification head (SVM-RBF), and the attribute training dataset (WEG^t) and vary the backbone architectures. We explore the effect of backbone size with different ResNets, ranging from iResNet18 (IR18) [16] to ResNet101 (RN101) [28], as well as ViT-T and ViT-S, with the InsightFace implementation[3]. We then focus on specific aspects in the ablation study (model training strategy, classification head type, attribute classification dataset size, and structure). We detail the experimental setup in Appendix D.

4.2. Main results

Table 1 compares REGI with seven competitive baselines. The results show that REGI variants improve ethnicity prediction across all datasets and evaluation dimensions, and are second-best after the much larger QWEN3 models for gender. We discuss results in more detail below.

Ethnicity. REGI outperforms existing baselines trained specifically for the attribute for all audited dimensions and datasets. Accuracy gains range from 7 to 16 points rela-

[2]Not to be confounded with the original DeepFace publication [70]
[3]https://github.com/deepinsight/insightface

	Arch.	#GFLOP	Ethnicity								Gender					
			Accuracy ↑			Fairness ↓			Robust. ↓		Accuracy ↑		Fairness ↓		Robust. ↓	
			BFW	RFW	WEG^t	BFW	RFW	WEG^t	BFW	WEG^t	BFW	WEG^t	BFW	WEG^t	BFW	WEG^t
FFNet$_{DS}$	RN34	1.94	87.9	88.1	74.4	3.50	4.10	6.44	0.19	0.29	96.3	91.4	2.01	0.9	0.13	0.28
DeepFace$_{DS}$	VGG16	31.06	82.3	82.3	71.4	1.72	3.15	2.96	0.29	0.37	85.4	74.8	26.9	47.4	0.31	0.33
FaceXF$_{FT}$	Swin-B	30.92	84.7	83.4	67.2	3.54	3.11	8.25	0.23	0.33	95.1	89.3	0.97	4.64	0.12	0.21
CLIP$_{FT}$	ViT-B	8.74	83.3	83.3	70.7	5.23	3.37	5.14	0.21	0.36	73.3	79.2	53.2	39.8	0.44	0.42
CLIP$_{FT}$	ViT-L	155.4	87.6	89.2	80.8	2.83	1.60	3.20	0.20	0.25	68.0	75.1	63.9	48.3	0.38	0.39
QWEN3$_{FT}$	4B	1101	93.6	92.8	83.9	2.28	2.40	5.15	0.11	0.19	98.2	95.4	0.87	2.10	0.07	0.16
QWEN3$_{FT}$	8B	8767	89.7	90.9	80.3	4.82	3.81	7.45	0.16	0.18	**99.1**	**95.5**	1.65	2.30	**0.05**	**0.15**
REGI	RN18	0.98	94.0	93.4	86.2	1.34	1.93	2.93	0.11	0.22	96.9	91.9	0.86	0.82	0.11	0.26
REGI	RN34	1.94	92.3	91.3	82.8	1.48	2.53	3.29	0.15	0.27	95.9	89.8	1.15	3.02	0.15	0.31
REGI	IR18	5.24	94.6	94.5	88.2	1.13	1.52	2.51	0.10	0.19	97.3	93.4	0.95	0.44	0.10	0.22
REGI	RN50	2.18	95.5	95.0	88.6	1.01	1.43	2.82	0.08	0.17	97.8	93.7	0.32	1.33	0.08	0.21
REGI	ViT-T	2.76	95.5	94.7	88.9	0.89	1.39	2.72	0.09	0.17	97.6	93.9	1.00	0.88	0.08	0.20
REGI	RN101	4.04	95.4	95.0	88.9	1.08	1.37	2.84	0.08	0.17	97.9	94.1	0.74	1.02	0.08	0.20
REGI	IR50	12.64	95.9	**95.8**	**90.8**	0.95	1.22	**2.52**	**0.07**	**0.13**	97.9	94.7	1.14	**0.11**	0.07	0.19
REGI	ViT-S	11.00	**96.2**	95.3	89.9	**0.82**	**1.20**	2.68	**0.07**	0.15	97.9	94.5	**0.31**	1.74	0.08	0.19

Table 1. Comparison of ethnicity and gender inference based on accuracy, fairness, and robustness dimensions. We evaluate REGI with different backbones and a fixed pretraining dataset (1.4M images), our siversified dataset (WEG^{tr}), and a classification head (SVM-RBF) against competitive baselines. Note that QWEN3's architecture column reflects parameter count (4B/8B) rather than a standard architecture acronym.

	WEG^t				BFW				RFW			
Training	**FC**	**LIN**	**MLP**	**RBF**	**FC**	**LIN**	**MLP**	**RBF**	**FC**	**LIN**	**MLP**	**RBF**
FairFace	70.1	75.2	71.7	*75.3*	79.2	83.7	82.6	*85.9*	80.1	85.3	82.4	*86.2*
BUPT	81.7	81.7	84.0	*87.9*	90.5	90.6	93.7	*95.6*	90.4	90.6	93.2	*95.2*
WEG^{tr}	**83.1**	**84.1**	**89.2**	***90.8***	**91.0**	**91.6**	**95.5**	***95.9***	**90.9**	**91.3**	**94.8**	***95.8***

Table 2. Impact of the classification head and of the ethnicity training dataset. *Best results for the classification head.* **Best results for the ethnicity training dataset.**

tive to FairFace, with larger gains for DeepFace and FaceXFormer. Fairness and robustness scores are at least halved in all three cases. Interestingly, the differences remain significant even when using compact backbones, such as RN34, the architecture used by FairFace. REGI modeling choices are different from those of the FFNet, DeepFace, and FaceXFormer. We test the effect of the training dataset, pretraining dataset, and classification head separately in the ablation experiments. The gains are also significant for the four generalist baselines that employ much larger architectures and pretraining datasets but do not explicitly focus on ethnicity. QWEN3 with 4B parameters comes closest to REGI, but still lags behind by 2 to 7 accuracy points, doubles the fairness metric, and is roughly 50% higher on robustness. Intriguingly, the larger QWEN3 version performs worse than the 4B version. Efficiency-wise, QWEN3 inference is orders of magnitude slower than that of REGI, a difference that is important for practical deployment. Performance differences between compact and larger models remain modest when using a pretrained FR model with an RBF classifier,highlighting the reliability and efficiency of this simple yet effective DAI pipeline. In practice, we recommend RN50 or ViT-T for constrained environments and IR50 for optimal performance. Globally, the ethnicity results highlight that this attribute remains challenging to predict, and REGI outperforms all baselines, while being much more efficient than large VLMs.

Gender. This inference is simpler than ethnicity inference, as previously noted [34] and confirmed by results in Tab. 1. REGI again outperforms the three baselines trained specifically for gender prediction and the two CLIP versions. The QWEN3 models improve accuracy by 1 point compared with the best REGI versions, are more robust, but remain less fair. This behavior might result from gender information being more available than ethnicity information in the QWEN3 pretraining data. However, the small accuracy gains come at the cost of much worse efficiency, as illustrated in the GFLOPS column of Table 1. REGI offers a good balance between performance and efficiency for gender prediction.

Test dataset. We provide a complete description of the specifications of WEG^t, the contributed test dataset, as a benchmark set in App. B. Results show that WEG^t is more challenging than BFW and RFW for both attributes and the three evaluation dimensions. The remaining error rate is roughly twice as high for ethnicity and nearly five times higher for gender. This behavior is explained by the emphasis put on diversification during its curation. Consequently, WEG^t constitutes a valuable resource for ethnicity and gender prediction evaluation, as explained in App. B.

4.3. Ablation study

We conduct additional experiments to assess the specific effects of the newly proposed pipeline components. We focus on ethnicity, the more challenging of the two modeled attributes. They examine the roles of the selected classification head, the task-related training dataset, the model training strategy, and the pretraining size. Unless indicated otherwise, we fix the backbone (IR50) and the pretraining dataset (WebFace with 1.4M samples). We present further ablation experiments in Appendix F.

Classification head effect. In addition to the SVM with an RBF kernel used in the main experiment, we test: (1) FC - the standard fully-connected linear layer used in deep neural networks; (2) LIN - linear SVM head with a different optimization objective compared with (1); (3) MLP - non-linear MLP head using two 512-dimensional layers followed by a ReLU layer, the head architecture recommended for age prediction in [50]. The results in Table 2 confirm that the RBF head provides the best performance, followed by the MLP, the linear SVM, and FC. This finding confirms our theoretical analysis regarding the high compatibility between an RBF layer and the pretrained angular encoder.

Ethnicity training dataset. We reuse the pretraining configuration to conduct a head-to-head comparison with the FairFace, BUPT, and WEG^t test datasets. The results in Table 2 show that WEG^{tr} outperforms BUPT for all test datasets and classification heads. The margin is larger for WEG^t, the most challenging test datasets. WEG and BUPT have similar sizes and include the same number of entities. The main difference lies in the increased emphasis on fine-

	90K	180K	360K	700K	1.4M	4M
WEG^t	82.3	85.4	87.6	89.4	90.8	**91.2**
BFW	91.5	93.6	94.6	95.2	95.9	**96.2**
RFW	91.0	92.9	94.1	95.2	**95.8**	95.6

(a) Impact of face recognition pretraining dataset size on ethnicity accuracy (%) with the RBF head trained on WEG^{tr}. Last column is the model trained by [36] on Webface 4M.

	E2E	FT	RBF
WEG^t	78.1	89.6	**90.8**
BFW	87.0	95.8	**95.9**
RFW	87.7	95.2	**95.8**

(b) Impact of training strategies for ethnicity accuracy (%). FT and RBF pretraining uses 1.4M images. WEG^{tr} is used for training.

grained ethnic diversity in WEG. The performance obtained with the publicly available FairFace dataset is significantly lower. This finding underscores the advantage of carefully composing the training datasets. Note that the slight difference in size does not explain the performance gap since WEG^t clearly outperforms the FaceFace dataset even when its size is smaller (see Appendix F).

Pretraining size effect. The scores obtained by varying the size of the pretraining dataset in Table 3a indicate that performance increases for larger pretraining. 1.4M, but saturates beyond, probably as an effect of WebFace ethnicity imbalance when using the full dataset. The "larger is better" hypothesis, usually made for pretraining datasets, is only verified up to a point in our experiment.

Model training strategy. We compare the three model training strategies from Figure 2: end-to-end (E2E), fine-tuning (FT), and probing (RBF). E2E and FT were carefully optimized with Optuna. The results in Table 3b show that probing outperforms fine-tuning, in addition to being faster to train. The comparison of the first columns from Tables 3a and 3b shows that probing clearly outperforms end-to-end training even when pretraining is done with a small dataset. This last result refutes the hypothesis that REGI performance gains arise solely from strong pretraining. We conclude that face recognition is a suitable pretraining task for ethnicity and gender inference and, advantageously, replaces end-to-end training.

5. Limitations and Ethics

Model size and pretraining data: larger may still be better. Our results show that transfer learning from compact face recognition models yields strong performance. We intently focus on compact models to favor evaluation thoroughness with limited computational resources (Embedded AI). However, we observe modest but consistent gains when using larger architectures and more extensive pretraining datasets. This suggests the need for choosing a tradeoff between performances and computational requirement.

Limited interpretability of learned demographic features. While our approach improves performance across accuracy, fairness, and robustness, the internal representations learned by deep models remain opaque. This lack of interpretability hampers transparency in high-stakes applications, making it difficult to audit which features influence predictions. Improving interpretability without sacrificing accuracy remains an important research direction, particularly as current architectures prioritize representational abstraction over explainability.

Demographic attribute completeness. We focus on ethnicity and gender because these attributes are stable at the identity level. As such, they can be reliably labeled from existing face recognition datasets by aggregating predictions for individual samples. Age varies within each identity, and aggregation is not possible. The challenges of estimating this attribute are highlighted in [50], and the authors recommend using their solution, which consists of strong pretraining followed by non-linear probing with an MLP. Their finding for age strongly resonates with ours for ethnicity and gender.

Ethical risks and potential misuse. We designed the pipeline for fairness auditing and scientific reproducibility. However, it could be repurposed for unethical uses such as surveillance, profiling, or discrimination. The ability to infer sensitive demographic attributes from facial images at scale raises significant concerns around consent, privacy, and bias amplification. While releasing models and code promotes transparency and peer scrutiny, technical safeguards alone are insufficient. A broader dialogue is needed within the research, ethics, and policy communities to define acceptable use cases, auditing frameworks, and accountability mechanisms.

6. Conclusion

We revisit the process of inferring ethnicity and gender and propose diverse training and test sets for these tasks. We implement a transfer-learning–based pipeline that combines a model pretrained for face recognition and a non-linear classification head, supported by a theoretical motivation. We extend the evaluation framework beyond accuracy and fairness by introducing robustness as a third critical dimension. We thoroughly evaluate the proposed approach and demonstrate consistent performance for ethnicity, the more challenging attribute, and an interesting performance-efficiency balance for gender.

Looking ahead, we will evaluate the effects of larger pretrained models while preserving the emphasis on dataset diversity. We will study embedding interpretability techniques to further improve the transparency of the demographic attributes inference. Finally, we will investigate ways to automatically label diversified age datasets to also address this demographic attribute.

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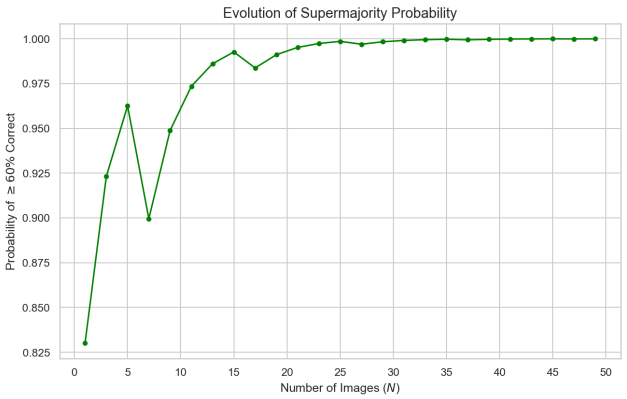


Figure 3. Theoretical evolution of the supermajority label being accurate depending on N, when k = 0.6 N (i.e., our threshold for WEG ethnicity estimation), and with p = 0.83

A. Balanced Sampling via Iterative Proportional Fitting

In this section, we formally describe the label inference and the sampling for creating WEG^{tr}, as mentioned in subsection 3.1, and validate it experimentally. The motivation behind the sampling is to create a dataset that faithfully reproduces the original structure of fine ethnicities, which are not compartmentalized into the 4 coarse ethnicities, but can often overlap them. We identify a good approach, a sampling based on the conditional probability $p(g|e)$ of a coarse ethnicity g, given a fine ethnicity e.

A.1. Theoretical and Experimental Guarantees for Label Inference Through Identity-Wise Majority Voting

Leveraging the predictions made by an ethnicity or gender classifier on several images of the same identity allows the aggregated label to substantially exceed the per-image accuracy of the underlying model. Formally, let p denote the probability that the classifier assigns the correct label to a single image. Assuming that prediction errors across different images of the same identity are approximately independent – a reasonable assumption given the diversity of pose, illumination, and expression within a WebFace42M identity – the number of correct predictions over N images follows a $\mathrm{Binomial}(N, p)$ distribution. We retain the majority label only when it represents at least 60% (the threshold we kept for ethnicity estimation) of the N predictions, i.e., when the number of correct votes $k \geq \lceil 0.6\,N \rceil$. The probability that this supermajority label is correct is therefore:

$$P(k \geq \lceil 0.6\,N \rceil) \;=\; \sum_{k=\lceil 0.6\,N \rceil}^{N} \binom{N}{k} p^k\,(1-p)^{N-k}. \quad (5)$$

As N increases, this probability converges rapidly to 1 for any $p > 0.6$. We provide the theoretical evolution of this probability for $p = 0.83$ (FFNet accuracy on the full BUPT dataset) in Figure 3. Even at $N = 5$ images, the supermajority probability already reaches 96.3%, and at $N = 20$ it exceeds 99.6%. In practice, our filtering step retains only identities whose per-image consensus exceeds a stricter threshold, further tightening this bound. The obtained label accuracy on BUPT with our thresholds reaches 95.5%, vastly outperforming the per-image baseline and reducing the error rate well below 5%, thereby breaking the potential circularity induced by the use of an automatic annotator.

As a final quality check, we manually verify the accuracy of ethnicity and gender labels for WEG^t. Confirming the robustness of the identity-level attribute labeling, fewer than 0.3% of samples are removed during manual verification.

A.2. Balanced Sampling Procedure

Consider a dataset with N groups indexed by $g \in \{1, \ldots, N\}$ and M subgroups indexed by $e \in \{1, \ldots, M\}$. Let $p(g|e)$ denote the conditional probability of group g given subgroup e, satisfying $\sum_{g=1}^{N} p(g|e) = 1$ for all e.

We propose the following two-stage sampling procedure:

1. Sample a group g uniformly with probability $p(g) = \frac{1}{N}$
2. Within group g, sample subgroup e with probability $p(e|g) = \frac{p(g|e)}{S_g}$

where $S_g = \sum_{e=1}^{M} p(g|e)$ denotes the total mass of group g across all subgroups.

Theorem 1 (Balanced Sampling Condition)**.** *The marginal sampling probability of subgroup e is*

$$p(e) = \frac{1}{N} \sum_{g=1}^{N} \frac{p(g|e)}{S_g} \quad (6)$$

When $p(g) = \frac{1}{N}$ the following equivalence yields: ***$p(e) = \frac{1}{M}$ for all e if and only if $S_g = \frac{M}{N}$ for all $g \in \{1, \ldots, N\}$. Thus, when sampling with regard to $p(g|e)$, ensuring that $S_g = \frac{M}{N}$ and sampling the coarse ethnicities uniformly enforce diversity with regard to the fine ethnicities.***

This theoretical element motivates the use of this sampling.

Proof. By the law of total probability,

$$p(e) = \sum_{g=1}^{N} p(e|g)p(g) = \sum_{g=1}^{N} \frac{p(g|e)}{S_g} \cdot \frac{1}{N} = \frac{1}{N} \sum_{g=1}^{N} \frac{p(g|e)}{S_g}. \quad (7)$$

For the sufficiency, if $S_g = \frac{M}{N}$ for all g, then

$$p(e) = \frac{1}{N}\sum_{g=1}^{N}\frac{p(g|e)}{M/N} = \frac{N}{M}\cdot\frac{1}{N}\sum_{g=1}^{N}p(g|e) = \frac{1}{M}. \quad (8)$$

For the necessity proof, suppose $p(e) = \frac{1}{M}$ for all e. Note that

$$S_g = \sum_{e=1}^{M} p(g|e) \quad (9)$$

$$= \sum_{e=1}^{M} p(e|g) * \underbrace{p(g)}_{1/N} / \underbrace{p(e)}_{1/M} \quad (10)$$

$$= \underbrace{\sum_{e=1}^{M} p(e|g)}_{=1} * M/N = \frac{M}{N} \quad (11)$$

If the distribution is uniform over subgroups while maintaining uniform group sampling, the balance constraint $S_g = \frac{M}{N}$ must hold to ensure compatibility of the marginals. □

A.3. Enforcing Balance via Iterative Proportional Fitting

Given an empirical conditional probability matrix $\{\tilde{P}_e\}_{e=1}^{M} = \{\tilde{p}(g|e)\}_{e=1}^{M}$, we seek an adjusted matrix $\{P_e^*\}_{e=1}^{M} = \{p^*(g|e)\}_{e=1}^{M}$ that satisfies the balance constraints while minimally perturbing the original distribution. This is formalized as a constrained Kullback-Leibler divergence minimization problem [12, 31].

Problem 1 (Constrained KL-Divergence Minimization)**.** *Find P^* that solves*

$$\begin{aligned} \min_{P_e} \sum_{e}^{M} \quad & D_{\mathrm{KL}}(P_e \| \tilde{P}_e) = \sum_{e=1}^{M}\sum_{g=1}^{N} p(g|e)\log\frac{p(g|e)}{\tilde{p}(g|e)} \\ \textit{s.t.} \quad & \sum_{e=1}^{M} p(g|e) = \frac{M}{N} \quad \forall g \in \{1,\dots,N\} \\ & \sum_{g=1}^{N} p(g|e) = 1 \quad \forall e \in \{1,\dots,M\} \\ & p(g|e) \geq 0 \quad \forall g, e \end{aligned} \quad (12)$$

The Iterative Proportional Fitting (IPF) algorithm [15, 67] provides an efficient method to solve Problem 1. IPF alternates between normalizing rows and columns until convergence. We apply this algorithm to our problem, following the pseudo code of Algorithm 1.

Algorithm 1 Iterative Proportional Fitting (IPF)

Input: Empirical matrix $\tilde{P}$, tolerance ϵ
Initialize: $P^{(0)} \leftarrow \tilde{P}$, $t \leftarrow 0$
repeat
  // Row normalization
  for $g = 1$ to N **do**
    $r_g^{(t)} \leftarrow \sum_{e=1}^{M} p^{(t)}(g|e)$
    $\hat{p}^{(t)}(g|e) \leftarrow p^{(t)}(g|e) \cdot \frac{M/N}{r_g^{(t)}} \quad \forall e$
  end for
  // Column normalization
  for $e = 1$ to M **do**
    $c_e^{(t)} \leftarrow \sum_{g=1}^{N} \hat{p}^{(t)}(g|e)$
    $p^{(t+1)}(g|e) \leftarrow \hat{p}^{(t)}(g|e) \cdot \frac{1}{c_e^{(t)}} \quad \forall g$
  end for
  $t \leftarrow t + 1$
until $\max_g |r_g^{(t)} - M/N| < \epsilon$ and $\max_e |c_e^{(t)} - 1| < \epsilon$
Return: $P^* \leftarrow P^{(t)}$

A.4. Theoretical Guarantees

Algorithm 1 converges to the unique solution P^* of Problem 1. The solution has the multiplicative scaling form

$$p^*(g|e) = u_g^* v_e^* \tilde{p}(g|e) \quad (13)$$

where $u_g^*, v_e^* > 0$ are scaling factors satisfying the marginal constraints

$$\sum_{e=1}^{M} u_g^* v_e^* \tilde{p}(g|e) = \frac{M}{N}, \quad \sum_{g=1}^{N} u_g^* v_e^* \tilde{p}(g|e) = 1. \quad (14)$$

The convergence and optimality of IPF for KL-divergence minimization subject to marginal constraints is well-established in the literature [12, 67]. The algorithm performs block coordinate ascent on the dual problem, where each row (column) normalization step corresponds to optimizing over one set of dual variables while fixing the other [53]. The objective function is strictly convex, ensuring a unique global optimum. By strong duality [8], the primal solution obtained from the optimal dual variables $p^*(g|e) = u_g^* v_e^* \tilde{p}(g|e)$ is the unique minimizer of Problem 1. For a comprehensive treatment of I-projections and the information-geometric perspective, we refer to [12].

The IPF solution P^* preserves the zero structure of $\tilde{P}$ (if $\tilde{p}(g|e) = 0$ then $p^*(g|e) = 0$) and minimally perturbs the original distribution in the sense of KL-divergence. This makes it particularly suitable for applications where the empirical associations $\tilde{p}(g|e)$ reflect meaningful semantic relationships that should be preserved while enforcing balance constraints.

A.5. Experimental Verification

We compare across 3 sampling strategies, the results obtained when training REGI on the resulting dataset. The 3 sampling strategies are as follow:

- **WEG Uniform** consists in sampling an ethnicity present in a coarse ethnicity g with probability $\frac{1}{M_g}$ with $M_g = \#\{e, e \in g\}$, the number of fine ethnicities present in g.
- **WEG Cond** consists in sampling with regard to $\tilde{p}(g|e)$.
- **WEG IPF** consists in sampling with regard to $p^*(g|e)$

We show in Table 4 that the WEG IPF sampling yields equivalent Accuracy (slightly better), but most importantly, yields fairer and more robust predictions than the other approaches. We think these results are a consequence of a more effective balancing (as shown by the Lorenz curve and the reported GINI coefficient in Figure 4) than other sampling strategies. Figure 4 Results are coherent with Theorem 1.

B. Comparison of WEG^t with Existing Benchmarks

In this section, we highlight the specific characteristics of WEG^t compared with existing benchmark sets. These proper characteristics can be decomposed in three points:
Fine-grained demographic diversity. WEG^t improves fine-grained demographic diversity over both BFW and RFW, as measured by the Gini coefficient (lower is better) computed across the fine-grained ethnicity sub-groups within each coarse label. WEG^t achieves a Gini coefficient of 0.35, compared with 0.48 for RFW and 0.49 for BFW. This lower value reflects a more balanced representation of fine-grained sub-groups within each coarse ethnic category, exposing identities that BFW and RFW systematically underrepresent. Concretely, the diversification procedure described in Section 3.1 — which samples uniformly over fine-grained ethnicities rather than only over the four coarse labels — directly reduces this imbalance and produces a test set that is more sensitive to intra-group performance variations.
Reduced accuracy saturation. BFW and RFW exhibit pronounced accuracy saturation for state-of-the-art models, which causes *fairness metrics to collapse*: when all groups are predicted with near-ceiling accuracy, the degree of bias (DoB) tends toward zero regardless of the model's actual discriminative properties, rendering the benchmark uninformative. As shown in Table 1, several baselines already reach accuracy plateaus on BFW and RFW, while WEG^t preserves a larger margin for discrimination across all evaluated models. This additional headroom makes WEG^t *more discriminative* for both current and future models, and ensures that fairness metrics remain meaningful as DAI performance continues to improve.
Stronger robustness evaluation. WEG^t contains approximately $\times 3$ more identities and $\times 2.5$ more images than BFW, providing the statistical depth required for reliable intra-identity consistency estimation via the HomE metric (Eq. 4). RFW, by contrast, does not provide identity-level annotations and therefore cannot support the same identity-level robustness analysis; accuracy and fairness are the only dimensions assessable on RFW, as noted in Section 4.1. The larger number of identities in WEG^t also reduces the variance of the HomE estimator, yielding more stable robustness measurements across demographic groups and making cross-model comparisons more reliable.

C. Complete Theoretical Justification for RBF-SVM on Angular Margin Embeddings

We provide a novel unified geometric and probabilistic framework that theoretically justifies the use of RBF kernel classifiers for transfer learning from face recognition models pretrained with angular margin losses to demographic attribute inference. While the individual mathematical components we build upon are established results from directional statistics [42] and metric learning [16, 63], **our contribution is, to our knowledge, the first end-to-end theoretical analysis connecting (i) von Mises-Fisher(vMF)-structured pretraining, (ii) vMF kernel similarity, and (iii) vMF mixture decision boundaries in a unified framework for this transfer learning scenario**. Specifically, we characterize RBF-SVM decision functions as comparisons between von Mises-Fisher mixture models—an interpretation that has not appeared in prior work and provides principled guidance for architectural choices for DAI in fairness-aware face recognition.

C.1. Background: Angular Margin Losses and Hyperspherical Geometry

Modern face recognition models employ angular margin losses such as Cosface [75], ArcFace [16] or AdaFace [36], which operate on L2-normalized feature embeddings. Let $f : \mathbb{R}^{H\times W\times 3} \to \mathbb{R}^d$ denote a deep neural network that maps an input face image to a d-dimensional embedding. The normalization constraint ensures that feature vectors lie on the unit hypersphere:

$$\mathbf{z} = \frac{f(\mathbf{x})}{\|f(\mathbf{x})\|_2} \in \mathbb{S}^{d-1} := \{\mathbf{v} \in \mathbb{R}^d : \|\mathbf{v}\|_2 = 1\}. \quad (15)$$

Angular losses for a training sample $(\mathbf{x}_i, y_i)$ with ground-truth identity $y_i \in \{1, \dots, C\}$ can be written as:

$$\mathcal{L}_{\text{angular}} = -\log \frac{\exp(s\cos(\theta_{y_i} + m))}{\exp(s\cos(\theta_{y_i} + m)) + \sum_{j\neq y_i} \exp(s\cos\theta_j)}, \quad (16)$$

where $\theta_j = \arccos(\mathbf{w}_j^\top \mathbf{z}_i)$ is the angle between the embedding $\mathbf{z}_i$ and the j-th class center $\mathbf{w}_j \in \mathbb{S}^{d-1}$, s is a scale

	Arch.	#GFLOP	Ethnicity								Gender					
			Accuracy ↑			Fairness ↓			Robust. ↓		Accuracy ↑		Fairness ↓		Robust. ↓	
			BFW	*RFW*	*WEG^t*	*BFW*	*RFW*	*WEG^t*	*BFW*	*WEG^t*	*BFW*	*WEG^t*	*BFW*	*WEG^t*	*BFW*	*WEG^t*
`WEG IPF`	IR50	12.64	**95.9**	**95.8**	90.8	**0.95**	1.22	**2.52**	0.07	**0.13**	**97.9**	94.7	1.14	**0.11**	**0.07**	0.19
`WEG Cond`	IR50	12.64	95.8	**95.8**	90.7	1.21	**1.18**	2.66	0.07	0.14	97.8	94.7	1.28	0.41	0.08	**0.18**
`WEG Uniform`	IR50	12.64	95.6	95.6	**91.0**	1.12	1.23	2.61	0.07	0.14	**97.9**	94.7	**1.08**	0.29	**0.07**	0.19

Table 4. Ethnicity inference accuracy, fairness (DoB $\times 100$), and robustness (HoE) for three training set variants of REGI (IR50, SVM-RBF). Gender columns are left blank. **Bold**: best; underline: second best.

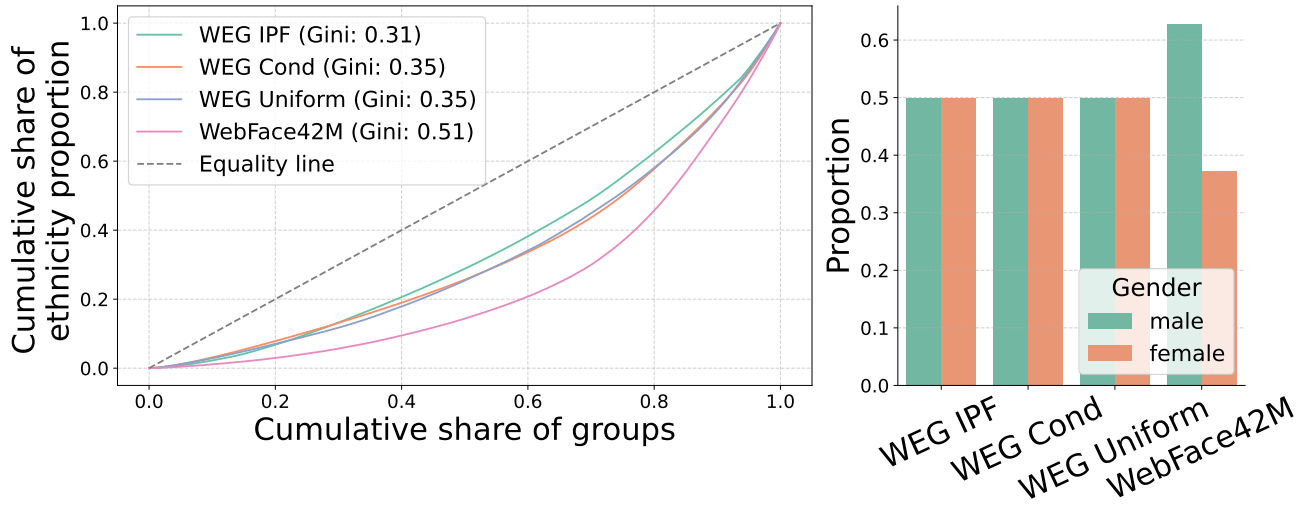


Figure 4. Fine ethnicities and gender proportion study for different balancing approaches.

parameter, and m is the angular margin penalty.

C.2. Established Connection: Angular Losses as von Mises-Fisher Distributions

The von Mises-Fisher (vMF) distribution is the canonical probability distribution for directional data on the unit hypersphere [42]. For a random unit vector $\mathbf{z} \in \mathbb{S}^{d-1}$, the vMF density is:

$$p(\mathbf{z} \mid \boldsymbol{\mu}, \kappa) = C_d(\kappa) \exp\left(\kappa \boldsymbol{\mu}^\top \mathbf{z}\right), \tag{17}$$

where $\boldsymbol{\mu} \in \mathbb{S}^{d-1}$ is the mean direction, $\kappa \geq 0$ is the concentration parameter, and $C_d(\kappa) = \kappa^{d/2-1}/[(2\pi)^{d/2} I_{d/2-1}(\kappa)]$ is the normalization constant with I_ν denoting the modified Bessel function of the first kind.

It is well-known [42, 63] that the softmax classifier with L2-normalized features and class centers is equivalent to a generative model where each class follows a vMF distribution. Specifically, under uniform class priors, the softmax probability (without angular margin) can be derived via Bayes' theorem:

$$P(y = c \mid \mathbf{z}) = \frac{p(\mathbf{z} \mid \mathbf{w}_c, s)}{\sum_{j=1}^{C} p(\mathbf{z} \mid \mathbf{w}_j, s)} = \frac{\exp(s \mathbf{w}_c^\top \mathbf{z})}{\sum_{j=1}^{C} \exp(s \mathbf{w}_j^\top \mathbf{z})}, \tag{18}$$

where each class c is modeled as a vMF distribution with mean direction $\mathbf{w}_c$ and shared concentration parameter $\kappa = s$. The normalization constants $C_d(s)$ cancel in the ratio.

The angular margin m modifies this by effectively decreasing the likelihood of the correct class during training through the transformation $\cos(\theta_c) \to \cos(\theta_c + m)$, forcing the model to achieve higher concentration (smaller intra-class angles) to overcome the penalty. This results in compact, well-separated vMF-like clusters on $\mathbb{S}^{d-1}$ after convergence.

C.3. Established Connection: RBF Kernel as von Mises-Fisher Similarity

The standard RBF (Gaussian) kernel between two vectors is:

$$K_{\text{RBF}}(\mathbf{z}_i, \mathbf{z}_j) = \exp\left(-\gamma \|\mathbf{z}_i - \mathbf{z}_j\|_2^2\right). \tag{19}$$

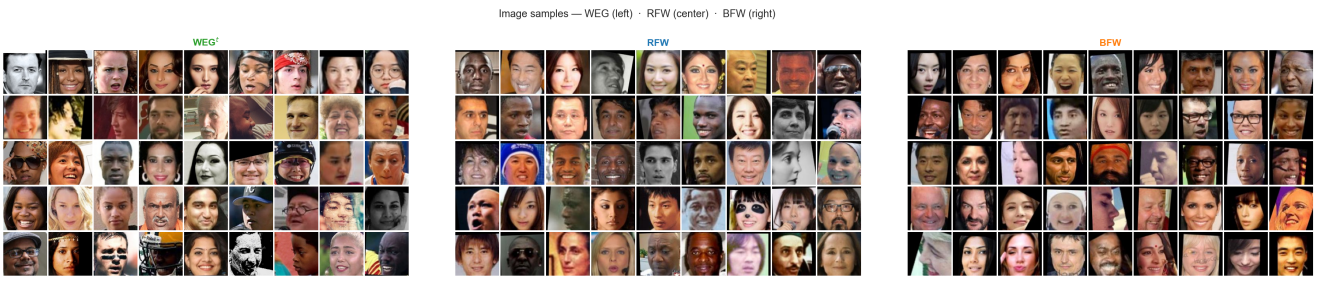


Figure 5. Samples drawn randomly from the different benchmark sets.

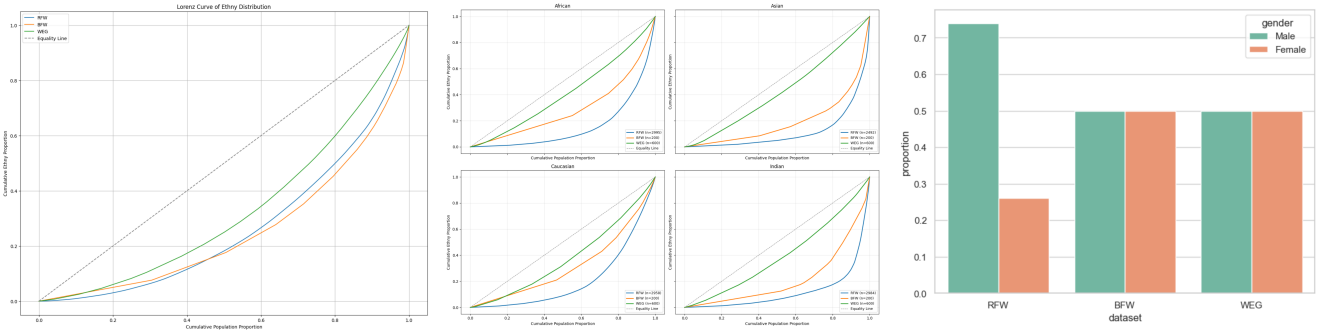


Figure 6. **Left** Lorenz curve for the overall distribution of fine ethnicities in different benchmark sets. The closer to the equality line, the more balanced.

Center: Lorenz curves for each coarse ethnicity. **Right**: Proportion occupied by each gender in each benchmark set.

For L2-normalized vectors $\mathbf{z}_i, \mathbf{z}_j \in \mathbb{S}^{d-1}$, a standard calculation yields:

$$\|\mathbf{z}_i - \mathbf{z}_j\|_2^2 = \mathbf{z}_i^\top \mathbf{z}_i - 2\mathbf{z}_i^\top \mathbf{z}_j + \mathbf{z}_j^\top \mathbf{z}_j = 2(1 - \mathbf{z}_i^\top \mathbf{z}_j). \tag{20}$$

Substituting into Eq. (19) and defining $\kappa_{\text{eff}} := 2\gamma$:

$$K_{\text{RBF}}(\mathbf{z}_i, \mathbf{z}_j) = \exp(-\kappa_{\text{eff}}) \cdot \exp(\kappa_{\text{eff}} \mathbf{z}_i^\top \mathbf{z}_j). \tag{21}$$

From the vMF density Eq. (17), setting $\boldsymbol{\mu} = \mathbf{z}_i$:

$$p(\mathbf{z}_j \mid \mathbf{z}_i, \kappa_{\text{eff}}) = C_d(\kappa_{\text{eff}}) \exp(\kappa_{\text{eff}} \mathbf{z}_i^\top \mathbf{z}_j). \tag{22}$$

Therefore, the RBF kernel satisfies:

$$K_{\text{RBF}}(\mathbf{z}_i, \mathbf{z}_j) = \frac{\exp(-\kappa_{\text{eff}})}{C_d(\kappa_{\text{eff}})} \cdot p(\mathbf{z}_j \mid \mathbf{z}_i, \kappa_{\text{eff}}) \propto p(\mathbf{z}_j \mid \mathbf{z}_i, \kappa_{\text{eff}}). \tag{23}$$

This establishes a classical result from directional statistics [42]: **the RBF kernel on normalized features computes a similarity measure proportional to the vMF likelihood**. The constant of proportionality does not affect the SVM decision boundary, as it scales all kernel evaluations uniformly.

C.4. Novel Contribution: Decision Boundaries as von Mises-Fisher Mixture Comparisons

We now present our key theoretical contribution: characterizing the RBF-SVM decision function as a comparison between vMF mixture models. This interpretation has not appeared in prior work and provides new insight into why RBF kernels are theoretically suited for this transfer learning scenario.

Consider binary demographic classification where $y_i \in \{-1, +1\}$ indicates class membership. The SVM decision function is:

$$f(\mathbf{z}) = \sum_{i \in \mathcal{S}} \alpha_i y_i K_{\text{RBF}}(\mathbf{z}_i, \mathbf{z}) + b, \tag{24}$$

where $\mathcal{S}$ is the set of support vector indices and $\alpha_i > 0$ are the dual coefficients.

Partitioning support vectors by class: $\mathcal{S}_+ := \{i \in \mathcal{S} : y_i = +1\}$ and $\mathcal{S}_- := \{i \in \mathcal{S} : y_i = -1\}$, we rewrite:

$$f(\mathbf{z}) = \sum_{i \in \mathcal{S}_+} \alpha_i K_{\text{RBF}}(\mathbf{z}_i, \mathbf{z}) - \sum_{i \in \mathcal{S}_-} \alpha_i K_{\text{RBF}}(\mathbf{z}_i, \mathbf{z}) + b. \tag{25}$$

Substituting the vMF equivalence from Eq. (23):

$$f(\mathbf{z}) = \Lambda \left[\sum_{i \in \mathcal{S}_+} \alpha_i p(\mathbf{z} \mid \mathbf{z}_i, \kappa_{\text{eff}}) - \sum_{i \in \mathcal{S}_-} \alpha_i p(\mathbf{z} \mid \mathbf{z}_i, \kappa_{\text{eff}}) \right] + b, \tag{26}$$

where $\Lambda := \exp(-\kappa_{\text{eff}})/C_d(\kappa_{\text{eff}})$ is a positive constant.

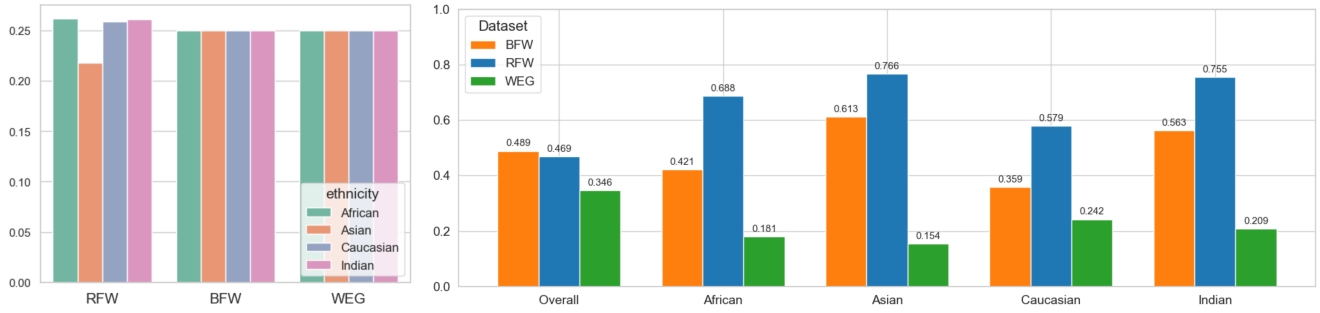


Figure 7. **Left:** Coarse ethnicities proportions for each benchmark set. **Right:** Gini coefficient for the fine ethnicities for each dataset. Overall stands for the Gini coefficient obtained on the whole dataset. We report the Gini coefficient of fine ethnicities for each coarse ethnicity.

Define normalized mixture weights:

$$w_i^+ := \frac{\alpha_i}{\sum_{j\in\mathcal{S}_+}\alpha_j} \quad \text{for } i \in \mathcal{S}_+, \quad \text{with} \sum_{i\in\mathcal{S}_+} w_i^+ = 1, \tag{27}$$

$$w_i^- := \frac{\alpha_i}{\sum_{j\in\mathcal{S}_-}\alpha_j} \quad \text{for } i \in \mathcal{S}_-, \quad \text{with} \sum_{i\in\mathcal{S}_-} w_i^- = 1. \tag{28}$$

Each sum in Eq. (26) can be interpreted as a **vMF mixture density**:

$$p_+(\mathbf{z}) := \sum_{i\in\mathcal{S}_+} w_i^+ p(\mathbf{z} \mid \mathbf{z}_i, \kappa_{\text{eff}}), \tag{29}$$

$$p_-(\mathbf{z}) := \sum_{i\in\mathcal{S}_-} w_i^- p(\mathbf{z} \mid \mathbf{z}_i, \kappa_{\text{eff}}). \tag{30}$$

These are well-defined mixture models where each support vector $\mathbf{z}_i$ serves as a vMF component center with concentration $\kappa_{\text{eff}} = 2\gamma$, and the normalized SVM coefficients $w_i^\pm$ act as mixture weights. The decision function becomes:

$$f(\mathbf{z}) = \Lambda\left[\beta_+ p_+(\mathbf{z}) - \beta_- p_-(\mathbf{z})\right] + b, \tag{31}$$

where $\beta_\pm := \sum_{j\in\mathcal{S}_\pm}\alpha_j$ represent the total mass assigned to each class.

Log-likelihood ratio interpretation. The classification decision $\text{sign}(f(\mathbf{z}))$ is equivalent to testing:

$$\hat{y} = \text{sign}\left[\beta_+ p_+(\mathbf{z}) - \beta_- p_-(\mathbf{z}) + b/\Lambda\right]. \tag{32}$$

Under the KKT stationarity condition, balanced support vector contributions ($\beta_+ \approx \beta_-$) yield, and by absorbing the bias into a threshold, this approximates a **log-likelihood ratio test**:

$$\hat{y} = \text{sign}\left[\log\frac{p_+(\mathbf{z})}{p_-(\mathbf{z})}\right] = \text{sign}\left[\log p_+(\mathbf{z}) - \log p_-(\mathbf{z})\right], \tag{33}$$

which is the optimal Bayes classifier for choosing between two generative models.

C.5. Implications: Geometric Consistency for Transfer Learning

We recall here the implications of our analysis, as stated in subsection 3.2

1. **Pretraining with angular margin losses** (e.g., AdaFace) organizes face embeddings on the hypersphere such that same-identity samples form tight vMF-like clusters (high κ), while different identities are angularly separated—this is established [16, 63].
2. **RBF kernel SVMs** operating on these normalized embeddings implicitly leverage the vMF geometry through Eq. (23), providing non-linear decision boundaries that respect the spherical structure—the kernel-vMF connection is established [42], but its application to this transfer learning scenario is novel.
3. **Decision boundaries as vMF mixture comparisons** (Eqs. (29)–(33)) provide a new probabilistic interpretation: the RBF-SVM learns to discriminate between demographic attributes by comparing which vMF mixture (constructed from support vectors) better explains a test embedding—**this characterization is our novel contribution, and justifies the use of balanced training data**: The more imbalanced the data, the poorer vMF representation the model will have at its disposal

Theoretical justification for RBF over linear classifiers. Unlike linear classifiers that assume Euclidean separability, the RBF kernel naturally accommodates the curved geometry of $\mathbb{S}^{d-1}$ through its vMF structure. While hyperspherical kernels have been studied for SVM [81], prior work did not connect them to vMF-based pretraining or characterize decision boundaries as mixture models. Our framework provides the first principled justification for RBF-SVM specifically in the context of transfer learning from angular-loss-pretrained face recognition to demographic inference.

Parameter selection guidance. The bandwidth parameter γ controls the effective concentration $\kappa_{\text{eff}} = 2\gamma$, allowing the classifier to adapt to the intrinsic clustering scale induced during pretraining. Higher γ increases sensitivity to angular proximity, appropriate when pretraining has produced highly concentrated (small intra-class variance) clusters. This provides interpretable, geometry-motivated guidance for hyperparameter selection—a contrast to deep non-linear classifiers that introduce numerous architectural parameters without explicit geometric motivation.

D. Implementation detailed description

D.1. Existing Datasets References

FairFace [34] is described in detail in section 2.

BUPT [77] is a dataset of real images that can be found at this URL. We actually consider *BUPT-BalanceFace* that contains 1.3M images from 28k celebrities. Four labels are considered, namely *caucasian*, *indian*, *asian* and *african*. The dataset is balanced w.r.t the number of identity per class (7k per label) and approximately balanced in terms of number of images: 324k for *african*, 325k for *asian*, 326k for *caucasian* and 375k for *indian*.

WebFace [82] is a dataset of real images, containing 4M identities represented by 260M face images, which a part (4M identities and 42M face images) has been curated to result into Webface42M.

D.2. Seed Images Collecion

The list of ethnicities covers the different regions of the world to optimize the diversity of the created datasets. The initial list was curated by prompting GPT-5.1 to provide up to 50 most populous ethnic groups including at least 1 million persons for major world regions (Africa, Asia, Europe, North America, South America, Oceania). In addition, we ran specific queries to get up to 20 ethnicities for large multi-ethnic countries, such as India or Indonesia, to improve their coverage. For instance, the list includes items such as *Javanese, Malay, Sundanese, Batak* for *Indonesia*. Since there is no one-to-one mapping between ethnicities and regions/countries, we deduplicated the items and obtained a list of 270 ethnicities.

We then populated the ethnicities by prompting GPT-5.1 to get 100 notable person names for each gender using ``List up to 100 notable living ETHNICITY GENDER. Diversify the professions and occupations. Format the output in JSON, providing the name and the profession. Avoid duplicate names.'' The list formatting was correct, but some duplicate names or name concatenations appeared in the results, and we filtered them out. After filtering, the list included nearly 30,000 unique names, with a 45% ratio of females.

Next, we collected 60 images for each name using a SERP API service over the Bing search engine and used RetinaFace [17] to detect faces in the images and kept those whose width was at least 96 pixels. We kept the identities that included at least 30 detected faces. Due to name ambiguity and retrieval imperfections, the search results contained noisy images and required filtering. We implemented an automatic filtering method inspired by [10]. We extracted face embeddings using the AdaFace R100 model trained with WebFace12M [4]. Then we compute an identity prototype by averaging the embeddings of the first five images including a single detected face. We compared each face embedding with the prototype identity representation and retained only faces that are closer to the target prototype than to any other. Finally, we applied a face verification protocol to further ensure that all faces belonged to the same identity. Preselected images of an identity were kept if their distance to at least one other image of the target identity was below a verification threshold. This threshold was optimized on the FAVCI2D dataset [54], a demographically diversified face verification dataset.

To verify the quality of the filtering, we manually annotated the images of 100 randomly-selected identities. The average accuracy of the face-identity attributions was 99.3%. This score confirms the robustness of the proposed seed identity filtering.

D.3. Training and Inference details

For training the end-to-end (E2E) and finetuned (FT) classifier (c.f. Table 3b, we set the training duration to 30 epochs for E2E and 10 epochs for FT, with a batch size of 256. We use a warmup-cosine learning rate scheduler. The learning rate reaches its peak at 1e-3 for E2E training and 1e-4 for FT. The whole backbone is updated in both cases. Fine-tuning is performed after the same 30-epoch face-recognition pretraining as REGI. For this ablation, use the same iResNet50 backbone for fair comparison. We use a CrossEntropy loss and the augmentation described below in both cases.

For training the **face recognition classifiers** (REGI pretraining task), we employed the standard setup from AdaFace. We used the AdaFace loss with parameter h set to 0.333 and m set to 0.4. We applied the standard framework augmentations for face recognition training: random flipping, crop augmentation, low-resolution augmentation, and photometric augmentation. We trained the model for 30 epochs using the SGD optimizer with momentum and weight decay fixed at 5e-4. The scheduler was a step LR scheduler that divides the learning rate by 10 at epochs 12, 20, and 24. To ensure stability of the AdaFace loss when training the ResNets (torchvision implementation) 50 and

[4] https://github.com/mk-minchul/adaface

101, we add a fully connected layer to the models to project the original 2024 latent space to 512.

The SVM implementation is from scikit-learn [52]. To accelerate the computation of cross distances, we implemented a wrapper that performs this phase on a GPU. This allows us to drastically reduce both training and inference time (training time drops from 40 minutes to 2 minutes on a training set of 100k samples). For the RBF kernel, we use the scaling gamma defined as $\frac{1}{D \cdot \mathrm{Var}(X)}$, where D is the dimension of the feature vectors (512) and X are the input vectors. For the ablation on the datasets (Table 2), we found that, in the case of imbalanced training set with regard to the predicted label, a weighted hinge loss yields better performance than the unweighted one. Thus, we use, when training on FairFace, and when training on BUPT for gender inference, the weighting defined as: $\mathrm{Weight}(\mathrm{Class}_i) = \frac{0.5}{\mathrm{Frequency}(\mathrm{Class}_i)}$. To yield full compatibility with the hypothesis of Equation 33, our dataset is carefully balanced with regard to gender and ethnicity (c.f. Appendix A), and thus does not require such weighting. We use the squared hinge loss with L2 penalization and 1.0 as the regularization parameter C.

All trainings are performed on a compute device having 250 GB of RAM, 32 CPU cores and an NVIDIA H100 GPU.

As for the inference of the VLMs we used the following approaches: For the CLIP models, we leverage the common text/image encoding space: we encode each class name and compute the cosine distance between the input image embedding and each class embedding, assigning the class with the smallest distance.

For QWEN models, we use the following prompt for ethnicity prediction: ``Please analyze the image and determine the apparent ancestry of the person shown. Answer with one of the following: "Asian", "Caucasian", "African", "Indian. Return only the ancestry label in lowercase, nothing else.''

As for gender: ``Please analyze the image and determine the apparent gender of the person shown. Answer with one of the following: 'male', 'female'. Return only the gender label in lowercase, nothing else.''

D.4. Architecture Parameters

We report in Table 5 the parameters and GFLOPS associated with each architecture mentioned in the main paper. While allowing for significantly better performances, the backbones used by REGI-Face require lower parameter count and computational power.

Architecture(s)	#Parameters (M)	GFLOPS
RN18	11.2	0.98
ViT-T	19.1	2.76
RN34	21.3	1.94
IR18	24.0	5.24
RN50	24.6	2.18
IR50	43.5	12.64
RN101	42.5	4.04
ViT-S	76.0	11.00
Swin-B	109	30.92
VGG-Face	134	31.06
CLIP ViT-L	427	155.4
CLIP ViT-B	151	8.74
QWEN3 4B	4437	1101
QWEN3 8B	8767	1973

Table 5. Parameter count and FLOPS (computed with ptflops, except for QWEN, which required the use of deepspeed.profiling) to architecture mapping for DAI pipelines.

E. Details on metrics

E.1. Performance Metrics

The annotation raw and group-wise performance of 1 is estimated with *Accuracy*, with the exception of the group-wise gender performance, which relies on the TPR, since it is a binary classification task, defined, with its counterpart FPR as:

- *True Match Rate (TMR)* or *True positive Rate (TPR)* or *Sensitivity* is the number of actual positive cases correctly identified over the actual number of positive samples in the dataset
- *False Match Rate (FMR)*, or *False Positive Rate (FPR)*, is the number of negative cases incorrectly identified as positive by the FAS over the actual number of negative samples.

E.2. Fairness Metrics

To estimate the fairness in the main paper, we considered the Degree of Bias (DoB) [22] for ethnicity inference, which is the standard deviation of the per-group accuracies. A lower standard deviation means that the performance varies less from one group to another, thus implying a fairer annotator.

Other aggregation metrics, such as *Demographic Parity Difference (DPD)* and *Demographic Parity Ratio (DPR)* [1, 2] are derived from the same principle, aiming at achieving equal representation of the demographic groups.

Equalized Odds Difference (EOD) and *Equalized Odds Ratio (EOR)* [1, 27] require that the FAS's performance is independent of the demographic group, that is, the TPR or FPR are the same across all groups. The EOD is defined by the absolute difference between the smallest and largest

values for the considered performance metric (TPR or FPR) across groups, while EOR is defined according to their ratio. Similarly to demographic parity, these metrics are in $[0, 1]$ a FAS is perfectly fair when its EOD is zero or its EOR is one. We used EOD for evaluating the fairness with regard to gender inference, which in a binary classification setup is equivalent as measuring the difference between the two TPR for Male and Female.

We report below additional plots to illustrate predictor fairness. Confirming the results reported in Table 1, Figure 8 reports the accuracy ethnicity-wise for the ethnicity prediction task, showing that REGI achieves stable prediction performance across all 4 segments. We also analyze the fairness of ethnicity prediction with regard to gender in Figure 9, highlighting the robustness of REGI.

E.3. Label Robustness Metric

Let us consider an image dataset $\mathcal{I} = \{I_j^{id}; id \in [\![1, D]\!], j \in [\![1, N_{id}]\!]\}$, with D identities, an identitity id being represented by N_{id} images. Such datasets are typically face recognition-like, containing images grouped by identity (BFW, WebFace, BUPT, CASIA, etc.). We consider a labeling function $f_{\mathcal{A}}^{\mathcal{C}}$ that attributes a label to each image, each label being an element of a finite set $\mathcal{C}$, and depends of a dataset $\mathcal{A}$ ($f_{\mathcal{A}}^{\mathcal{C}}$ is defined below). We propose to consider three metrics to reflect the robustness of the labeling system $(\mathcal{A}, \mathcal{C})$. The general idea is that a good labeling system should (at least) attribute as much as possible images of a given identity to the same label.

Macro majority accuracy. For each identity id, we consider the frequency of the majority label and compute the average over all labels of the dataset

$$MaMA = \frac{\sum_{id=1}^{D} \max_{k=1\dots N_{id}} \left(\sum_{j=1}^{N^{id}} \delta_{f_{\mathcal{A}}^{\mathcal{C}}(I_j^{id}) = f_{\mathcal{A}}^{\mathcal{C}}(I_k^{id})} \right)}{\sum_{id} N_{id}} \quad (34)$$

Micro majority accuracy For each identity, we consider the proportion of images labeled with the majority label and compute its average over all identities:

$$MiMA = \frac{1}{D} \sum_{id=1}^{D} \frac{1}{N_{id}} \max_{k=1\dots N_{id}} \left(\sum_{j=1}^{N^{id}} \delta_{f_{\mathcal{A}}^{\mathcal{C}}(I_j^{id}) = f_{\mathcal{A}}^{\mathcal{C}}(I_k^{id})} \right) \quad (35)$$

Last, let remind that in the main paper we consider, for each identity id, the relative frequency of each possible label $k \in \mathcal{C}$ noted as $p_{id}^k = \frac{1}{N_{id}} \sum_{j=1}^{N^{id}} \delta_{f_{\mathcal{A}}^{\mathcal{C}}(I_j^{id}) = f_{\mathcal{A}}^{\mathcal{C}}(I_k^{id})}$. We then compute the **homogeneity entropy** as the average entropy of p over all identities:

$$HomE = \frac{1}{D} \sum_{id} \frac{1}{\log(C)} \sum_{k} -p_{id}^k \log(p_{id}^k) \quad (4 \text{ revisited})$$

These metrics reflect the average variation of the attributions for each entity. Ideally, all images of a given entity are attributed to the same label. In this case, the *homogeneity entropy macro majority accuracy* and *micro majority accuracy* are respectively 0,1 and 1. In the worst case, the images of a given entity are uniformly distributed over all C labels, leading to the three metrics above being respectively 1, $\frac{1}{C}$ and $\frac{1}{C}$. We thus normalize MiMA and MaMA by multiplying by $\frac{C}{C-1}$ and removing $\frac{1}{C-1}$ such that the worst value is 0 and the best value is 1

Labelling function the function $f_{\mathcal{A}}^{\mathcal{C}}$ project an image onto a unique label in $\mathcal{C}$. It is defined according an image dataset $\mathcal{A}$. We consider two versions of $f_{\mathcal{A}}^{\mathcal{C}}$:

- a non-parametric function: each image of $\mathcal{A}$ is described with an image encoder that project it into an euclidean space. It then attributes a label to an image according to its nearest neighbors (1-NN or k-NN)
- a parametric function, such as a neural network, that is learnt on $\mathcal{A}$, with labels in $\mathcal{C}$, and a cross-entropy loss.

E.4. Robustness Results

Results of robustness for all the methods introduced in Table 1 are reported in Table 6, in terms of all three metrics. In this table, MiMA and MaMA show very similar results. This is not surprising. Their formulation implies different behavior under heterigeneous number of images per identity. To see their specific behavior, we reported robustness values on the IMDb dataset in Figure 14.

In this Figure, we compare REGI-Face against approaches based on skin color (CASCo, Beyond Skin Tone [73], and AI-Face hybrid annotation on IMDb [41]). Such a comparison between different taxonomies is not possible with traditional performance metrics, such as Accuracy, and is one of the strongest motivations for using our robustness metrics. While a perfect robustness score is not sufficient to assess that an annotator is perfect, **poor robustness metrics imply an annotation scheme of poor quality**. The application of these robustness analyses on skin tone-based approaches (Figure 13, Figure 12, Figure 11, and Figure 14) allows us to conclude that demographic schemes based on skin color are unfortunately of poor quality compared to classic 4-ethnicity classification. This is especially highlighted by the comparison between REGI Face and the hybrid annotation performed by [41] in Figure 14.

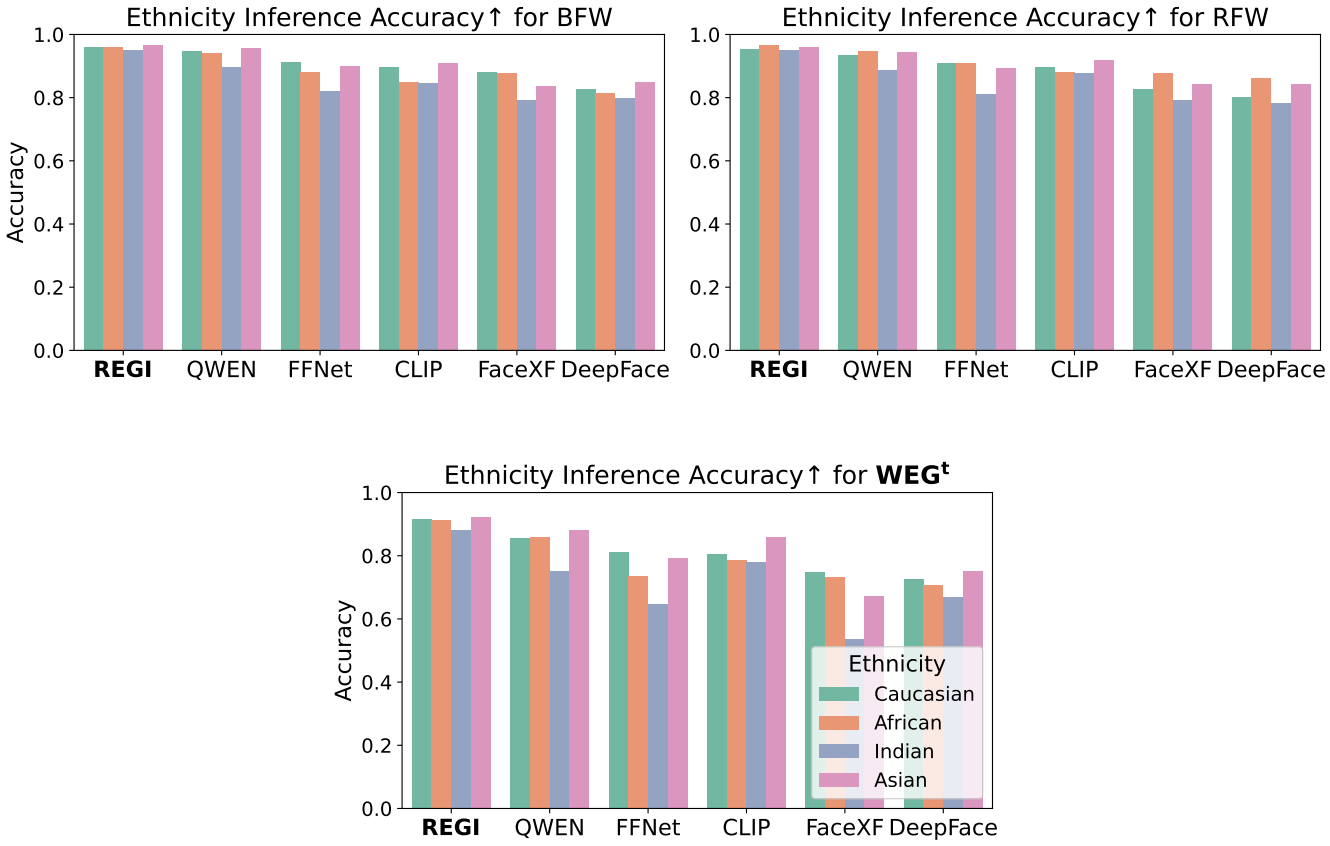


Figure 8. Accuracy per ethnicity group on ethnicity inference task.

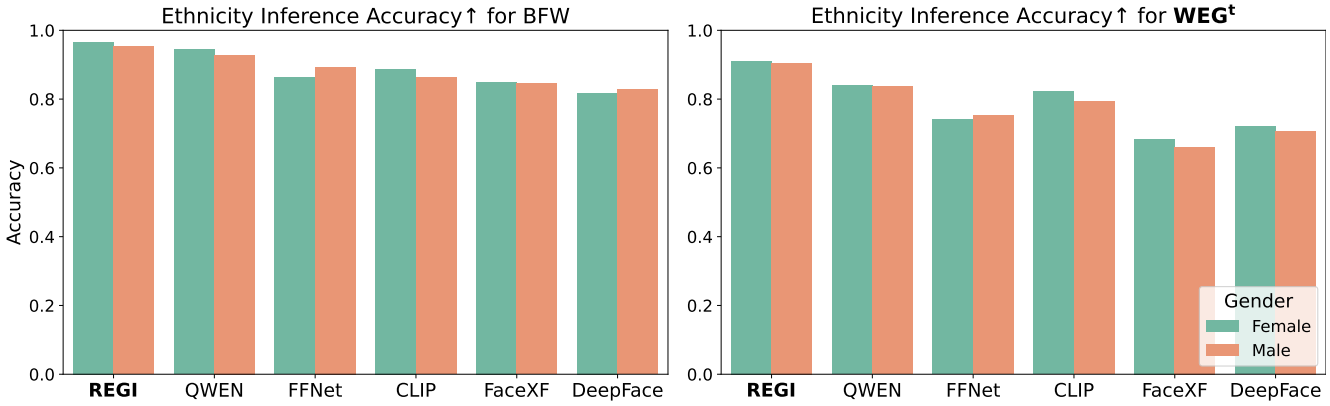


Figure 9. Accuracy per gender on ethnicity inference task.

F. Additional Abalation

F.1. Impact of the Demographic Training set structure

We perform an ablation on the training set structure fed to the SVM with RBF kernel in Table 7. The results clearly show that while the amount of data indeed affects REGI accuracy, ethnicity accuracy remains above several baselines, even under a scarce-data regime. Moreover, Table 7 shows that, when seeking more data, increasing the number of identities in the training set yields greater improvement than increasing the number of images per identity. This conclusion is similar to the one in subsection F.4

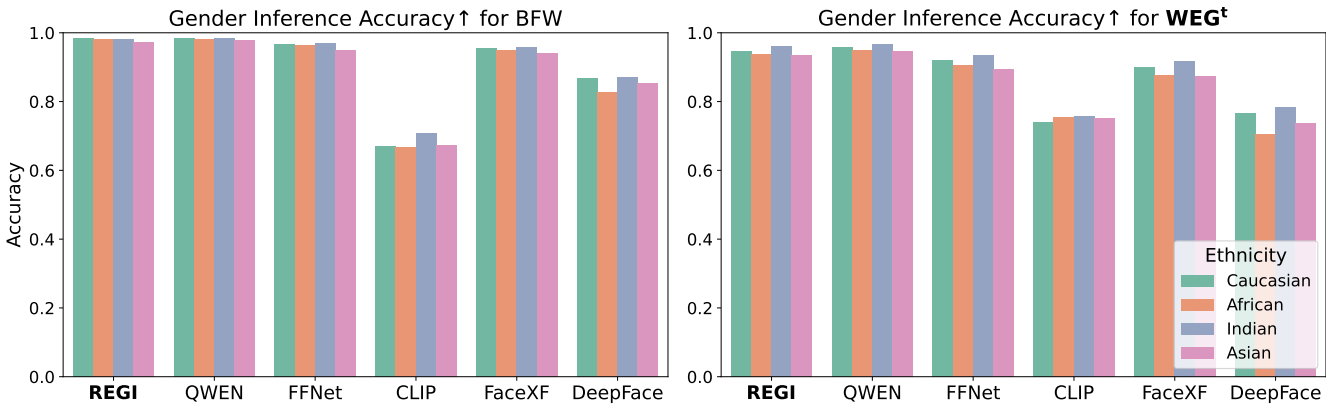


Figure 10. Accuracy per ethnicity on gender inference task.

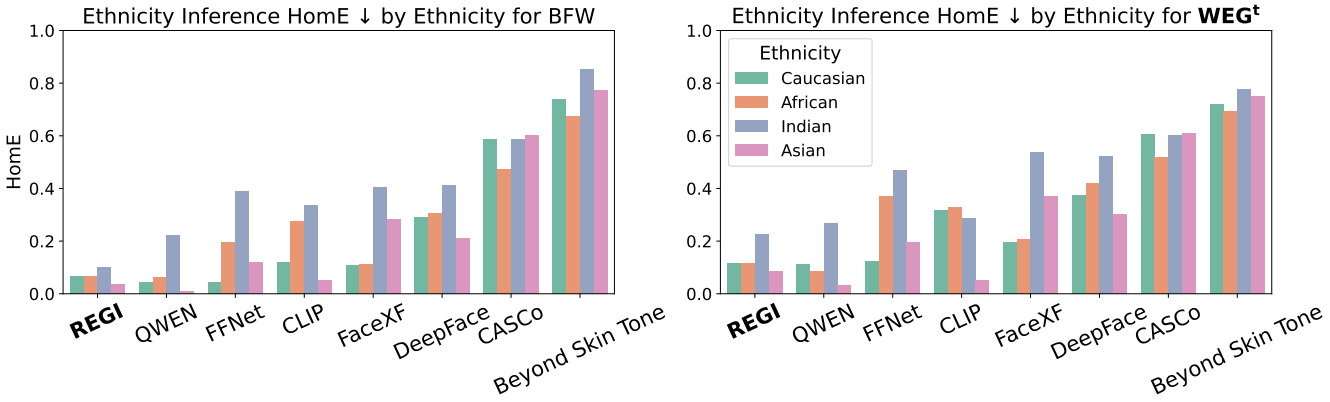


Figure 11. Per ethnicity segment robustness. We compare REGI with the baselines of Table 1 alongside two skin color based approaches.

F.2. Backbone Training impact on REGI

We further explore the impact of the face recognition backbone pretraining on REGI-Face.

More specifically, we plot the impact of the number of identities and images of the FR training set on the accuracy in Figure 15 and Figure 16. The figures show that, on both gender and ethnicity prediction tasks, a larger pretraining set yields better results, providing a **strong incentive for using the REGI approach**. While demographic annotations are limited and hard to obtain, face recognition data is abundant. These conclusions must, however, be tempered: The plot clearly shows a plateau effect. With the x scale in log2, doubling the dataset pretraining size indeed improves the final accuracy, but the benefits decrease as the training set size increases.

We extend the analysis of the backbone pretraining by exploring in Table 8 the impact of the balancing of the pretraining set. Surprisingly, a balanced training set seems to improve REGI-Face performance across all architectures.

Finally, to confirm the claim in subsection 3.2 on the compatibility of RBF-SVM with angular losses, we inspect the results on several FR angular loss, such as AdaFace, ArcFace, CurricularFace, and MagFace in Table 9. Confirming our claim, the performance of AdaFace, CurricularFace, and ArcFace is similar. The results on MagFace, which is farther from our theoretical hypothesis, tell a different story. On WEG^t, the accuracy drops by almost 3 points, demonstrating that using a loss that closely matches our framework is important.

F.3. Complementary results on Gender

We report in Table 10 the analysis on the impact of our training set WEG^{tr} and the SVM-RBFW head on the gender inference task. As stated in the previous section, we also report the impact of the number of identities in the pretraining set in Figure 16. We thus complete the results reported in Table 3a and Table 2. These results draw the same conclusion as the results on the ethnicity prediction task, confirming the superiority of REGI-Face, both in terms of the framework and the training set, over baselines.

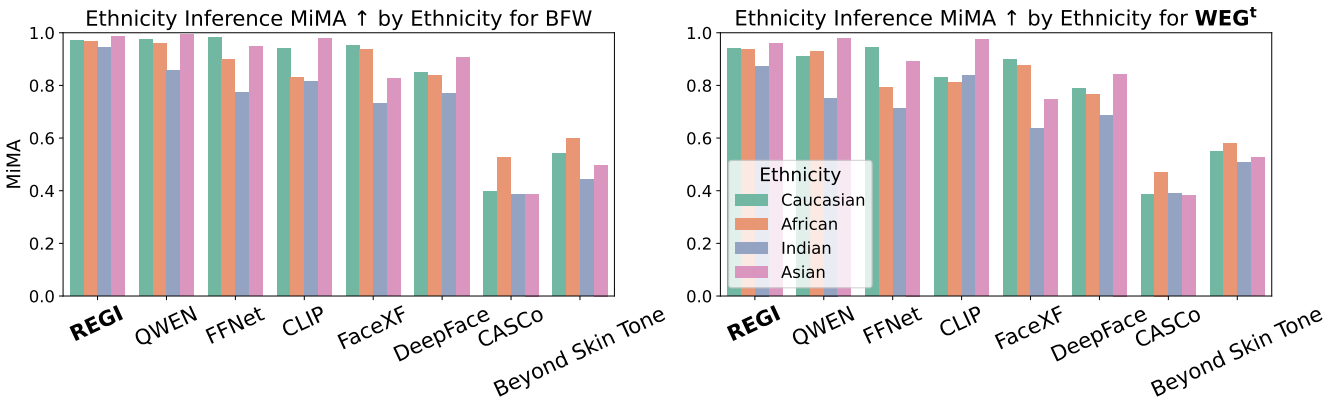


Figure 12. Per ethnicity segment robustness. We compare REGI with the baselines of Table 1 alongside two skin color based approaches.

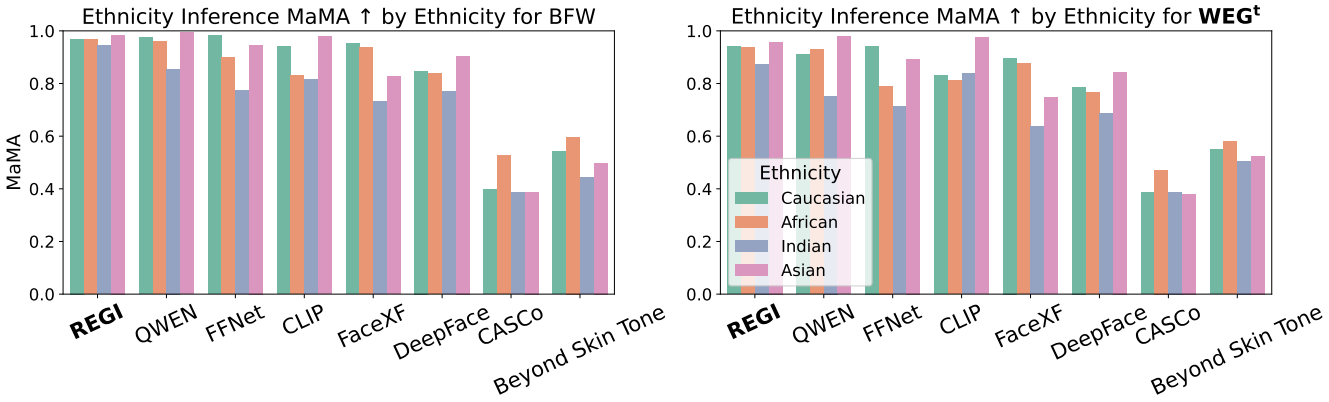


Figure 13. Per ethnicity segment robustness. We compare REGI with the baselines of Table 1 alongside two skin color based approaches.

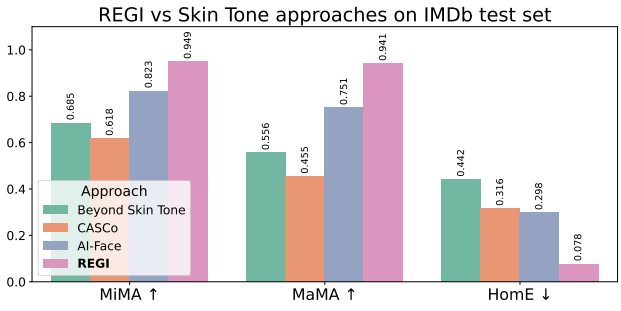


Figure 14. Robustness comparison between REGI and Skin Colors base approaches : Beyond Skin Tone (ICCV 2023), CASCo (Social Science Quarterly 2023) and the hybrid annotation of AI-Face (2025)

F.4. Marginal Effects Analysis

To rigorously quantify the impact of training configuration choices on DAI accuracy performance, we conduct marginal effects analysis using ordinary least squares (OLS) regression. This approach allows us to estimate the average causal effect of each variable while controlling for confounders, providing a more robust understanding than simple bivariate comparisons.

F.4.1. Methodology

Model Specification We specify two related regression models to analyze different aspects of our experimental results:

Model 1: Best Configuration Analysis For the first analysis, examining the best performing configuration per architecture, we model the target DAI accuracy metric y_i (best RFW validation accuracy at optimal epoch) as:

$$y_i = \beta_0 + \beta_1 \log(\text{IDs}_i) + \beta_2 \log(\text{Samples}_i) + \beta_3 \text{FV}_i + \beta_4 \text{Unbal}_i + \epsilon_i$$

where:

- IDs_i is the number of identities in the training set
- Samples_i is the number of samples per identity
- FV_i is the face verification accuracy of the base model
- Unbal_i is a binary indicator for unbalanced training (vs. balanced baseline)

	Arch.	#GFLOP	Ethnicity						Gender					
			MaMA ↑		MiMA ↑		HomE ↓		MaMA ↑		MiMA ↑		HomE ↓	
			BFW	WEG^t	BFW	WEG^t	BFW	WEG^t	BFW	WEG^t	BFW	WEG^t	BFW	WEG^t
FFNet [34]	RN34	1.94	89.97	83.51	90.00	83.51	0.19	0.29	96.60	91.97	96.60	91.97	0.13	0.28
DeepFace [64]	VGG16	31.06	83.15	79.12	83.16	79.12	0.30	0.37	90.00	89.41	90.01	89.41	0.31	0.33
FaceXF. [47]	Swin-B	30.92	86.18	78.93	86.19	78.93	0.23	0.33	95.64	90.50	95.63	90.50	0.12	0.21
CLIP [55]	ViT-B	8.74	88.29	78.88	88.34	78.88	0.21	0.36	82.93	84.90	82.99	84.90	0.44	0.42
CLIP [55]	ViT-L	155.4	89.17	87.88	89.16	88.13	0.20	0.21	86.29	86.32	86.34	86.32	0.39	0.39
QWEN3 [7]	4B	1101	94.60	89.44	94.59	89.44	0.11	0.19	98.22	**95.98**	98.22	**95.98**	0.07	**0.15**
QWEN3 [7]	8B	8767	91.54	89.93	91.58	89.93	0.16	0.18	**99.08**	**95.98**	**99.04**	**95.98**	**0.05**	**0.15**
REGI	RN18	0.98	94.48	88.34	94.48	88.34	0.11	0.22	97.25	92.42	97.25	92.42	0.11	0.26
REGI	RN34	1.94	92.85	85.09	92.85	85.09	0.15	0.27	96.11	90.61	96.12	90.61	0.15	0.31
REGI	IR18	5.24	95.31	89.93	95.31	89.93	0.10	0.19	97.61	93.85	97.62	93.85	0.10	0.22
REGI	RN50	2.18	96.02	90.60	96.01	90.60	0.08	0.17	98.11	94.17	98.12	94.17	0.08	0.21
REGI	ViT-T	2.76	96.00	90.75	96.01	90.75	0.09	0.17	97.95	94.36	97.97	94.36	0.08	0.20
REGI	RN101	4.04	96.07	91.02	96.06	91.02	0.08	0.17	98.21	94.39	98.22	94.39	0.08	0.20
REGI	IR50	12.64	**96.68**	**92.68**	**96.66**	**92.68**	**0.07**	**0.13**	98.18	94.84	98.18	94.84	0.07	0.19
REGI	ViT-S	11.00	96.63	91.69	96.63	91.69	**0.07**	0.15	98.17	94.80	98.18	94.80	0.08	0.19

Table 6. Comparison of ethnicity and gender inference based on MaMA, MiMA, and HomE dimensions. We evaluate REGI with different backbones and a fixed pretraining dataset (1.4M images), our diversified dataset (WEG^{tr}), and a classification head (SVM-RBF) against competitive baselines. Note that QWEN3's architecture column reflects parameter count (4B/8B) rather than a standard architecture acronym.

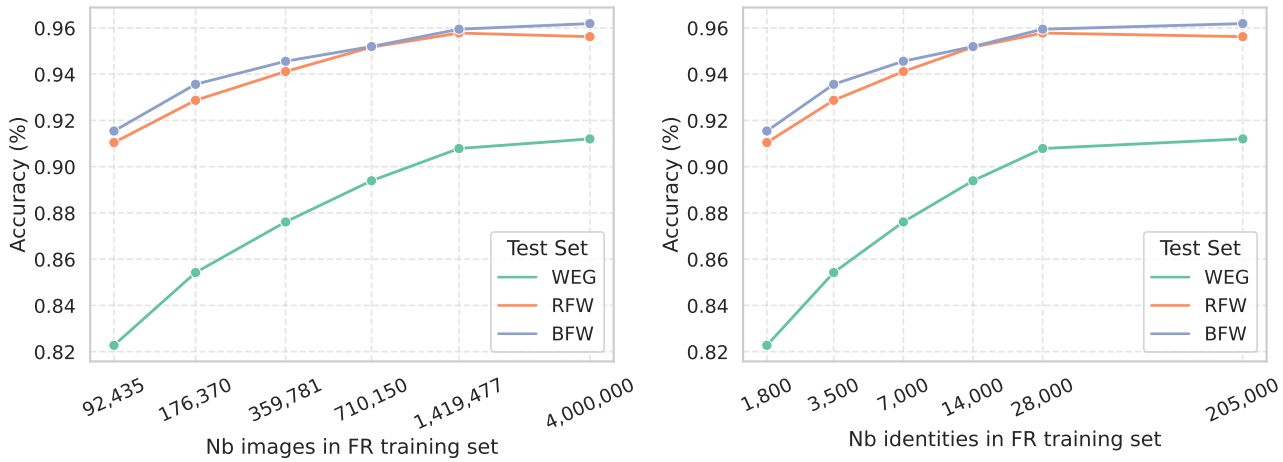


Figure 15. Impact of the size of the backbone's training set on REGI Face ethnicity accuracy. Results are plotted using an IR50 training on balanced subsets of WebFace. The predictor is following the configuration of REGI WEG^{tr} (except for the 4M checkpoint that is coming from AdaFace [36] repository)

- $\epsilon_i \sim \mathcal{N}(0, \sigma^2)$ is the error term

Model 2: Per-Epoch Analysis For the second analysis examining performance across training epochs for specific evaluation sets, we extend the model to:

$$y_{iet} = \beta_0 + \beta_1 \log(\text{IDs}_i) + \beta_2 \log(\text{Samples}_i) + \beta_3 f(\text{Epoch}_t) + \beta_4 C_i + \beta_5 \text{Unbal}_i + \epsilon_{iet}$$

where y_{iet} is the accuracy for configuration i, evaluation set e, at epoch t, and C_i represents either:

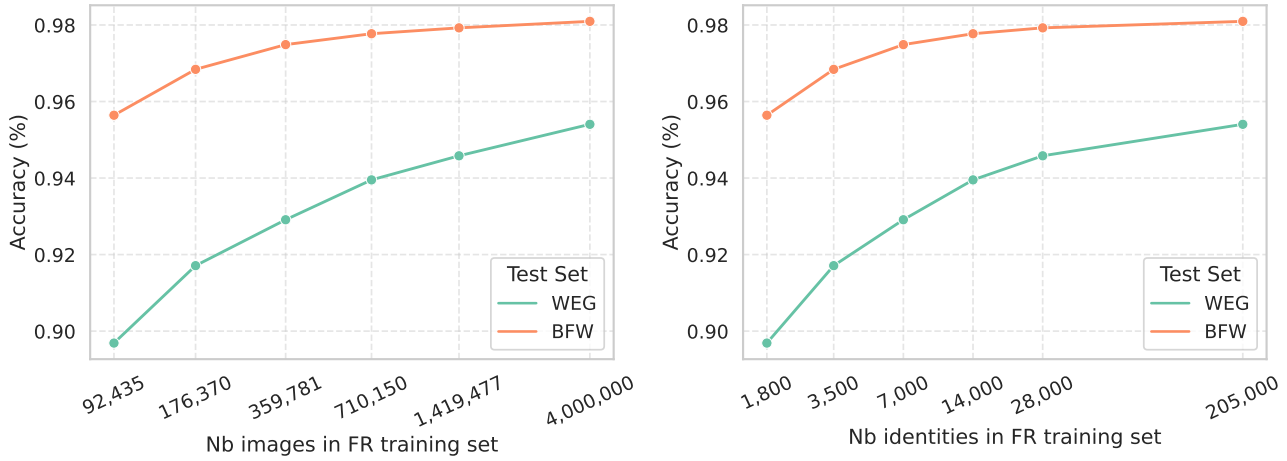


Figure 16. Impact of the size of the backbone's training set on REGI Face gender accuracy. Results are plotted using an IR50 training on balanced subsets of WebFace. The predictor is following the configuration of REGI WEG^{tr} (except for the 4M checkpoint that is coming from AdaFace [36] repository)

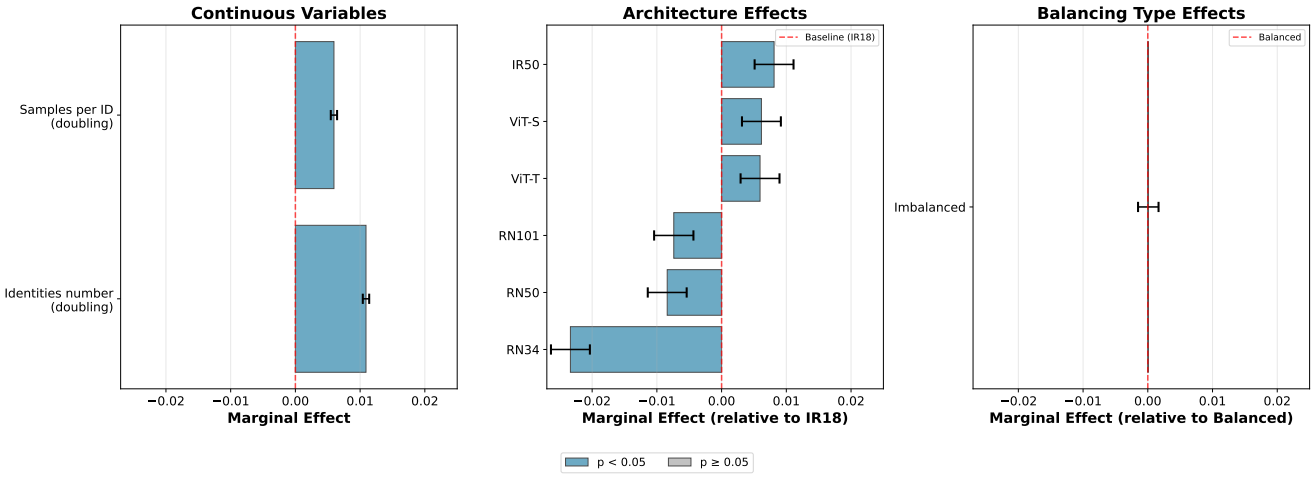


Figure 17. Marginal effects for OLS Model 2 on RFW

- A continuous control variable (FV accuracy on a specific benchmark), or
- A set of architecture dummy variables: $\sum_{k=2}^{K} \gamma_k \mathbb{1}(\text{Arch}_i = k)$

The function $f(\cdot)$ applies logarithmic transformation if epochs span more than one order of magnitude, otherwise uses linear scaling.

Variable Transformations To ensure proper model specification and interpretability, we apply the following transformations:

Log Transformations Variables spanning multiple orders of magnitude (number of training identities: 500–20,000; samples per identity: 1–40) are log-transformed. This addresses two concerns:

1. **Heteroskedasticity**: Raw values exhibit increasing variance with scale
2. **Interpretability**: Coefficients represent *elasticities* (percentage changes) rather than absolute changes

For a log-transformed variable, the marginal effect of doubling is computed as:

$$\Delta y_{\text{double}} = \beta_{\log(x)} \times \log(2) \approx 0.693 \beta_{\log(x)} \tag{36}$$

Categorical Variables Categorical variables (architecture, balancing type) are dummy-coded with one category as the baseline. The coefficient γ_k represents the average difference in outcome between category k and the baseline,

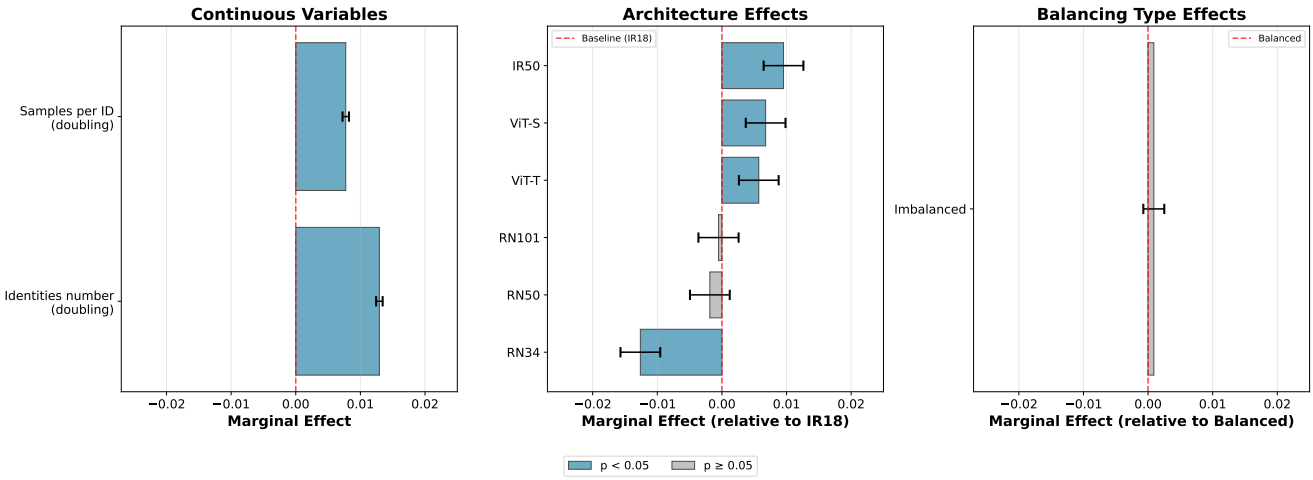


Figure 18. Marginal effects for OLS Model 2 on BFW

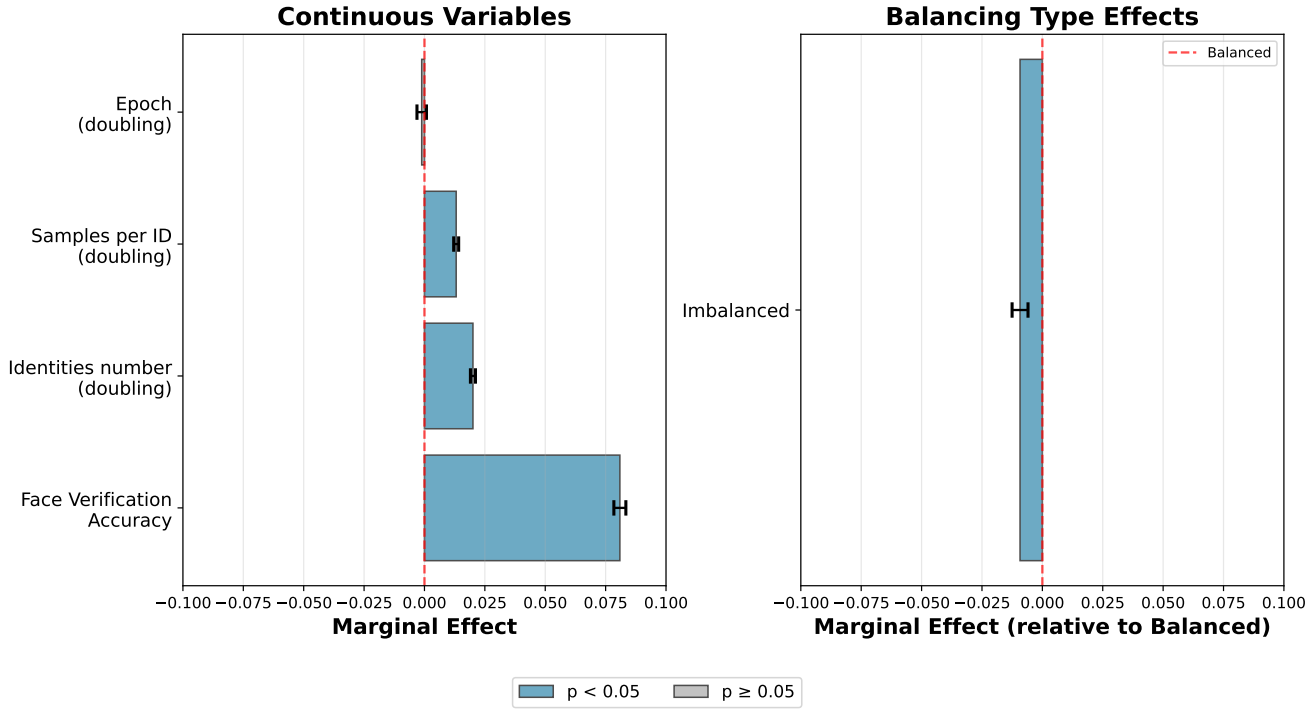


Figure 19. Marginal effects for OLS Model 1 on RFW

holding all other variables constant.

Why Control for Face Verification Accuracy? A critical methodological consideration is whether to include architecture as a categorical variable or control for face verification (FV) accuracy. We observe strong collinearity between architecture and FV performance , as different architectures inherently achieve different base recognition capabilities.

This creates two valid but distinct research questions:

- **Controlling for architecture**: "Which training configurations work best for each architecture?"
- **Controlling for FV accuracy**: "Given a certain base recognition quality, how do training choices affect DAI accuracy?"

We essentially consider Model 1 for concluding:

1. Architecture and FV accuracy measure fundamentally the same construct (model capability)

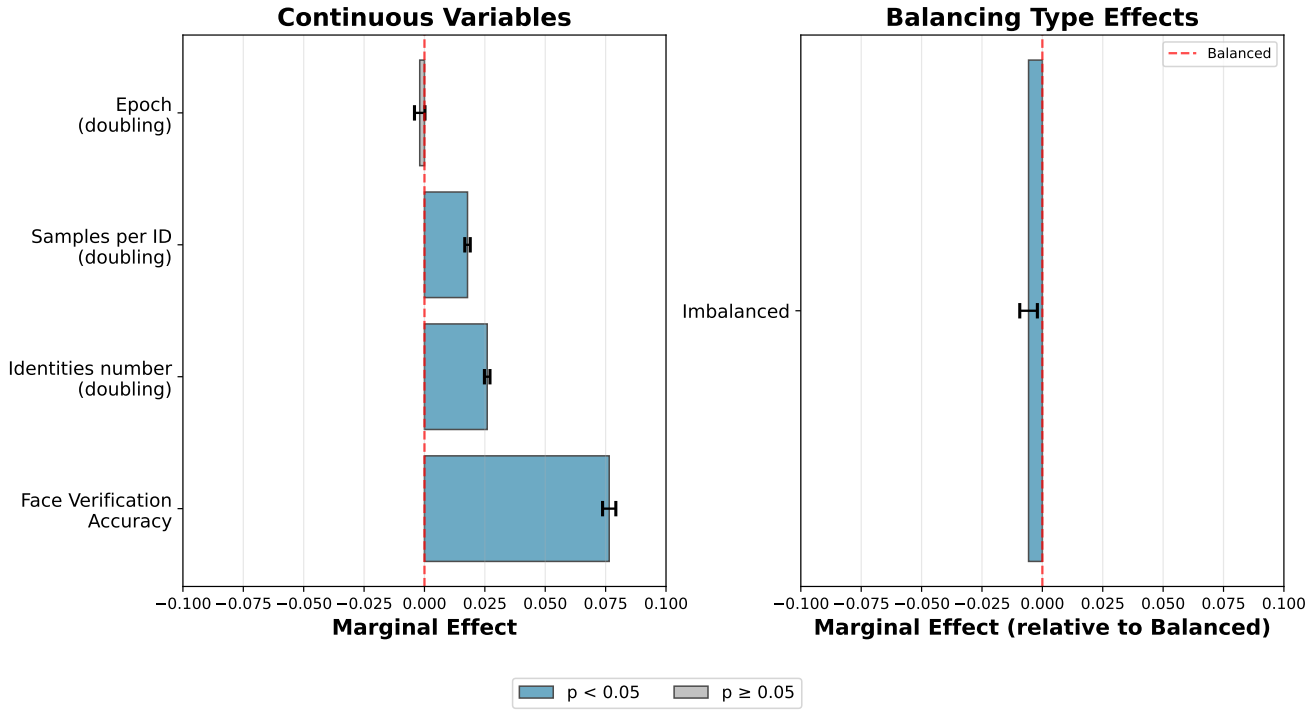


Figure 20. Marginal effects for OLS Model 1 on BFW

	1	2	5	10	15	20	40
500	79.0	80.9	82.2	82.9	83.1	83.2	83.5
1000	82.0	83.1	83.8	84.2	84.5	84.6	85.3
2000	83.5	84.2	84.8	85.6	87.1	87.6	88.8
5000	84.8	85.1	88.5	89.9	90.2	90.4	—
10000	85.6	88.4	90.5	91.0	—	—	—
20000	88.7	90.5	91.1	—	—	—	—

Table 7. Impact of the structure of the ethnicity classification dataset. We vary the number of identities (rows) and the number of samples per identity (columns). 40 is the maximum number of samples per identity. We limit the total dataset size to 100k. We report the WEG^t accuracy obtained with the IR50 backbone pretrained on 1M WebFace images.

Arch.	BFW		RFW		WEG^t	
	Imb	Bal	Imb	Bal	Imbb	Bal
IR- 18	94.6	**94.8**	94.5	**94.6**	88.2	**88.3**
IR- 50	95.8	**95.9**	95.3	**95.8**	89.9	**90.8**
RN101	94.8	**95.4**	94.4	**95.0**	87.9	**88.9**
RN34	91.0	**92.3**	89.8	**91.3**	80.9	**82.8**
RN50	94.6	**95.5**	94.4	**95.0**	87.9	**88.6**
ViT- s	95.3	**96.2**	94.9	**95.3**	88.9	**89.9**
ViT- t	94.9	**95.5**	94.2	**94.7**	88.4	**88.9**

Table 8. BFW, RFW, and WEG^t accuracy when pretraining the IR50 backbone with Imbalanced (Imb) or balanced (Bal) WebFace subset comprising 1M images.

2. Including both would introduce multicollinearity, inflating standard errors
3. FV accuracy provides a continuous, more granular measure of model quality
4. The coefficient on FV accuracy has a natural interpretation: DAI accuracy as a function of base recognition capability

For Model 2, we allow flexibility to examine either control strategy depending on the research question.

F.4.2. Results

Model 1: Optimal Configuration Analysis Figure 19 and Figure 20 present the marginal effects from Model 1. The regression achieves $R^2 = 0.856$ ($R^2_{\text{adj}} = 0.833$), indicating excellent model fit.

Training Set Size Doubling the number of training identities yields between **+1.21 and 2.25 percentage points** increase in DAI accuracy. This substantial effect demonstrates that DAI performance scales predictably with dataset diversity, consistent with prior work showing that represen-

Loss	Ethnicity								Gender					
	Accuracy ↑			Fairness ↓			Robust. ↓		Accuracy ↑		Fairness ↓		Robust. ↓	
	BFW	*RFW*	*WEG^t*	*BFW*	*RFW*	*WEG^t*	*BFW*	*WEG^t*	*BFW*	*WEG^t*	*BFW*	*WEG^t*	*BFW*	*WEG^t*
AdaFace	95.94	95.77	90.78	0.95	1.22	2.52	0.07	0.13	97.92	94.65	1.14	0.11	0.07	0.19
ArcFace	96.37	95.80	90.96	0.97	1.17	2.51	0.06	0.13	98.20	95.03	0.61	1.13	0.06	0.17
CurricularFace	95.83	95.52	90.35	1.02	1.27	2.35	0.07	0.14	97.96	95.12	0.73	0.52	0.07	0.17
MagFace	94.87	94.37	87.64	1.05	1.56	2.55	0.10	0.19	97.32	92.98	0.77	1.01	0.10	0.23

Table 9. Comparison of different losses based on ethnicity and gender dimensions (accuracy, fairness, robustness).

	WEG^t				BFW			
Training	**FC**	**LIN**	**MLP**	**RBF**	**FC**	**LIN**	**MLP**	**RBF**
FairFace	85.5	87.6	87.1	*89.2*	91.4	92.7	93.4	*93.6*
BUPT	87.7	88.7	92.7	*93.6*	94.5	94.7	97.3	*97.8*
WEG^{tr}	**88.9**	**89.5**	**94.3**	***94.7***	**94.9**	**95.3**	**97.6**	***97.9***

Table 10. Impact of the classification head and of the gender training dataset. *Best results for the classification head*. **Best results for the gender training dataset.**

tation matters for bias mitigation [9].

Samples per Identity Doubling samples per identity increases accuracy by **+0.64-1 percentage points**. While significant, this effect is roughly half that of adding identities, suggesting that *diversity* (more identities) is more valuable than *depth* (more samples per identity) for DAI accuracy.

Base Model Quality A 10-point increase in base FV accuracy corresponds to more than **+7.5 percentage point** improvement in DAI performance ($p < 0.001$). This is the strongest predictor in the model, indicating that DAI performance is highly sensitive to base recognition quality.

This finding has important implications: improving the underlying face verification model (through better architectures, pretraining, or optimization) yields larger DAI accuracy gains than dataset manipulation alone.

Balancing Strategy Unbalanced training shows a significant **negative** effect (between -0.5and -1 percentage points) compared to balanced training.

Model 2: Per-Epoch Analysis With the Model 2 Analysis, we confirm the performance results for the different backbones reported in Table 1. We also observe the same behavior for the impact of the number of identities and the number of samples per identity as with Model 1. Interestingly, though, the analysis on the balancing of the pretraining face recognition training set shows a non-significant effect, which contrasts with Model 1.

F.4.3. Marginal effects Conclusion

Marginal effects analysis reveals that the backbone choice (FV accuracy/architecture) is the strongest predictor of annotator performance, followed by training set diversity (number of identities) and depth (samples per identity).